Learning a Codebase

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April 20, 2026

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- 2.2 Time series sustained components touched by developer - 14 For developer day and components touched. . . . 60

3 Repository: 21 **62**
- 3.1 Time series sustained components touched by developer - 21 For developer commits and components touched. . . . 62
- 3.2 Time series sustained components touched by developer - 21 For developer day and components touched. . . . 65

1 Repository

1.1 Glossary - Summary

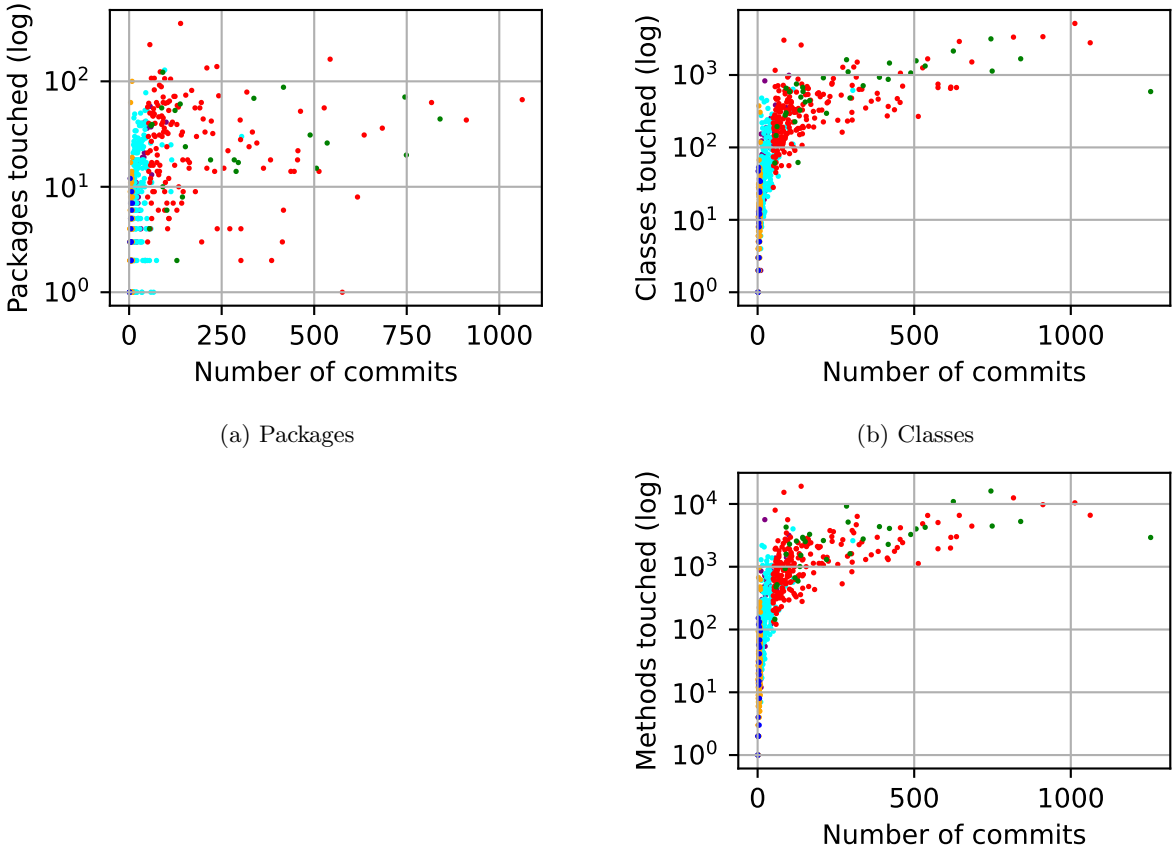
- The founder developer starts in the first six months of a project.
- Late joiner developers begin after six months.
- Transient developers have ten (10) or fewer commits.
- Moderate developers have more than ten (10) commits but fewer than 50 commits or commits for less than 250 days.
- Sustained developers make 50 or more commits and commit for 250 days or more.

1.2 Repository Summary Table

Table 1: Summary of fifteen open-source repositories identified from GitHub that have at least 1000 pull requests and at least three sustained late joiner developers. Sustained later joiner developers joining the project after six months and contributed at least 50 commits over a period of 250 days or more. Please note that five developers each worked on two repositories.

ID	Repo Name	Transient Founder	Moderate Founder	Sustained Founder	Transient Joiner	Moderate Joiner	Sustained Joiner	Commit	Start	End
3	activiti	4	7	6	47	20	5	8662	2010-Jun-18	On going
5	airbyte-platform	3	2	1	45	32	6	6784	2020-Aug-04	2023-Sep-27
7	ambari	0	3	0	34	32	15	4247	2011-Sep-22	On going
10	automq	2	0	0	40	10	3	1874	2011-Aug-10	2017-Apr-21
14	buck	8	5	6	119	85	21	14876	2013-Mar-21	2022-May-05
15	camel	0	4	1	257	63	13	36528	2007-Mar-19	On going
18	checkstyle	0	0	1	57	42	9	17338	2001-Jun-22	On going
20	cxf	6	6	3	27	17	3	7893	2008-Apr-29	On going
21	intellij-community	0	3	7	113	93	84	101247	2004-Nov-11	2018-Apr-18
26	guava	0	1	0	47	8	3	9685	2009-Sep-15	On going
28	jenkins	1	1	1	144	39	4	30252	2006-Nov-05	On going
33	openmrs-core	0	1	0	123	26	3	15757	2006-May-03	On going
36	presto	0	1	3	162	71	18	30605	2012-Aug-09	On going
37	quarkus	1	3	5	126	25	8	81612	2018-Jun-14	On going
39	selenium	3	3	0	55	17	8	23354	2004-Nov-03	On going
	Total	28	40	34	1396	580	203	390714		

1.3 Sample scatter touched against commit -



(a) Packages

(b) Classes

(c) Methods

Figure 1: Scatter plots of the log total number of components touched (y-axis) against the number of commits (x-axis) made by samples of developers capped at 203 from six (6) categories: Transient Founder (Blue, 28), Moderate Founder (Purple, 40), Sustained Founder (Green, 34), Transient Later Joiner (Orange, 203), Moderate Later Joiner (Cyan, 203), Sustained Later Joiner (Red, 203).

1.4 Total commits by developer - Developer Commits

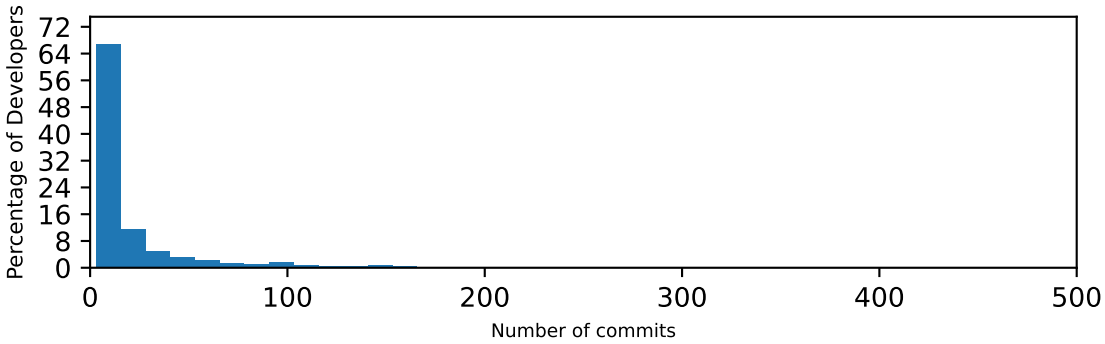
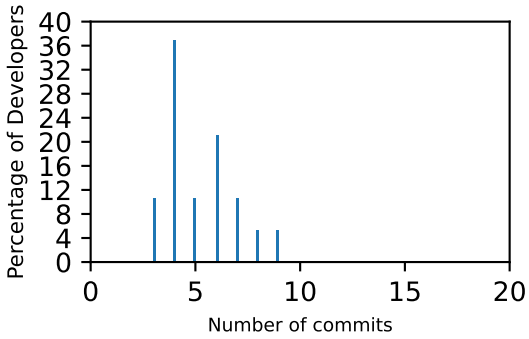
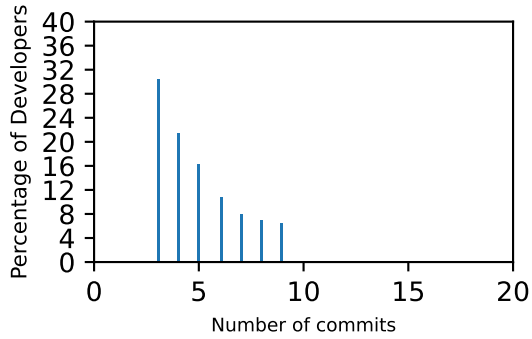


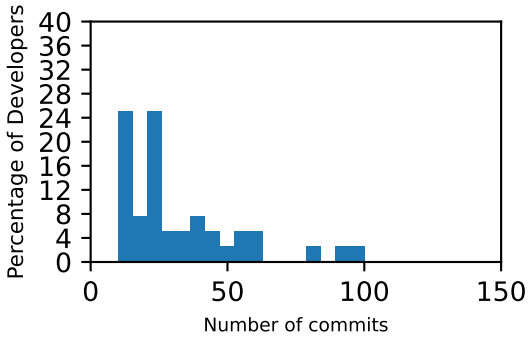
Figure 2: Histogram of the number from all developers (n=1906) (y-axis) against the total number of commits (x-axis) in 15 repositories sampled from GitHub.



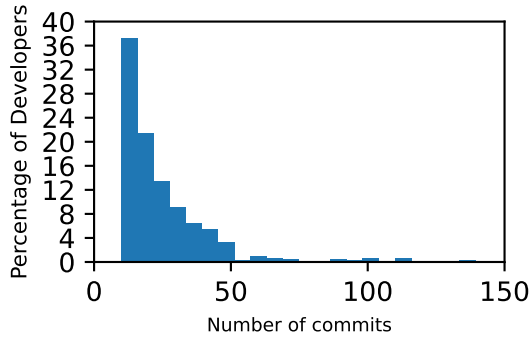
(a) Transient founder (n=19).



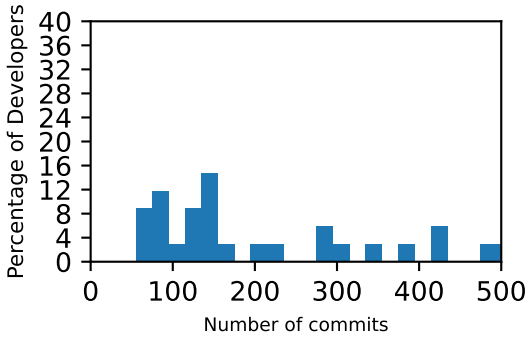
(b) Transient later joiner (n=1030).



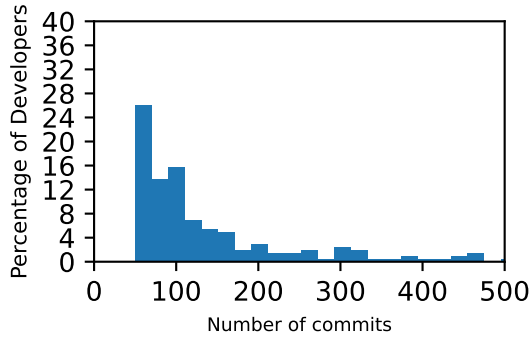
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).



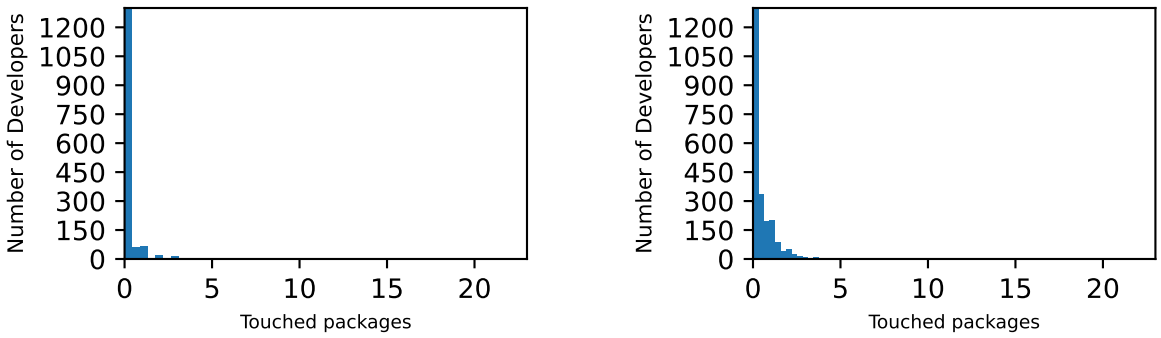
(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

Figure 3: Histogram of percentage of developers (y-axis) against total commits made (x-axis) in six (6) categories. This is from 15 repositories sampled from GitHub.

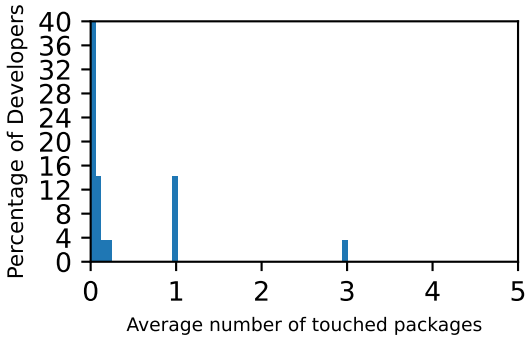
1.5 Histogram components touched by developer - For components with total



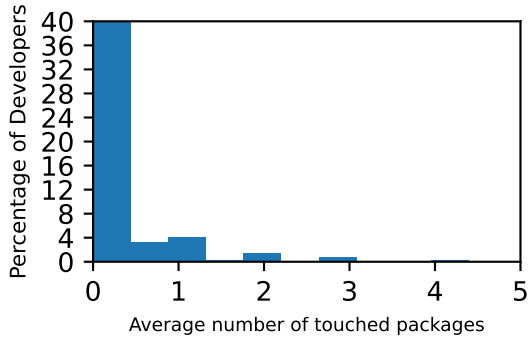
(a) By Day.

(b) By Commit.

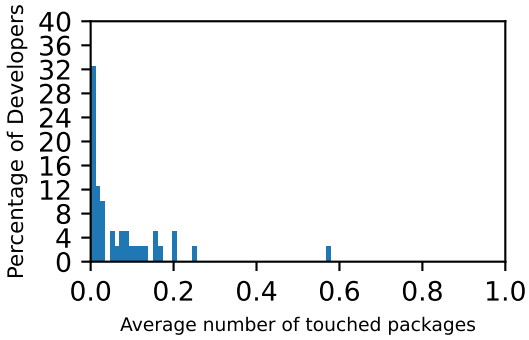
Figure 4: Histogram of the number of developers (y-axis) against the average packages touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



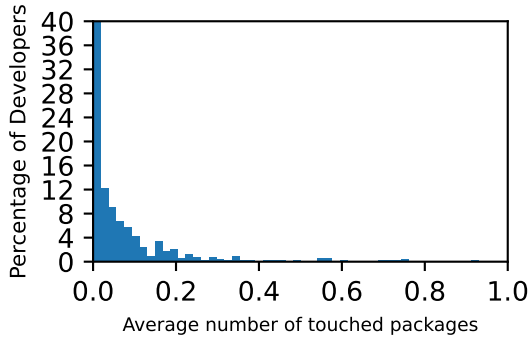
(a) Transient founder (n=28).



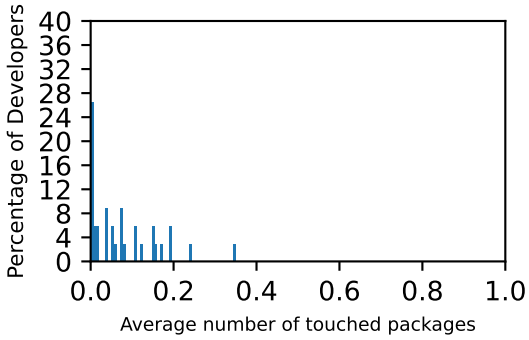
(b) Transient later joiner (n=1396).



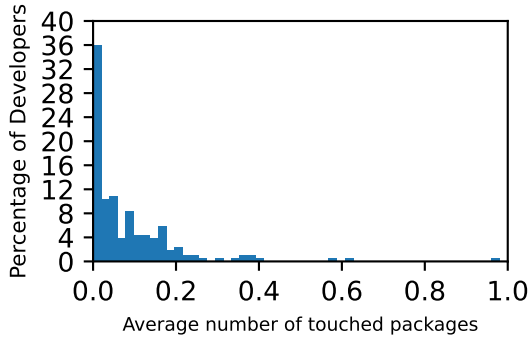
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

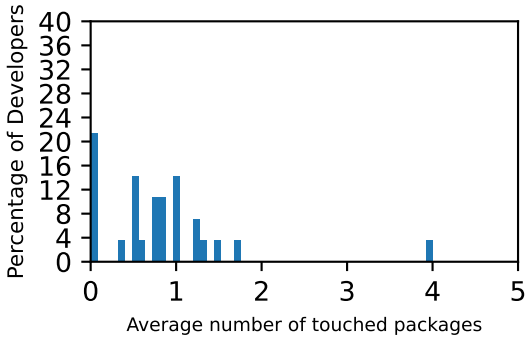


(e) Sustained founder (n=34).

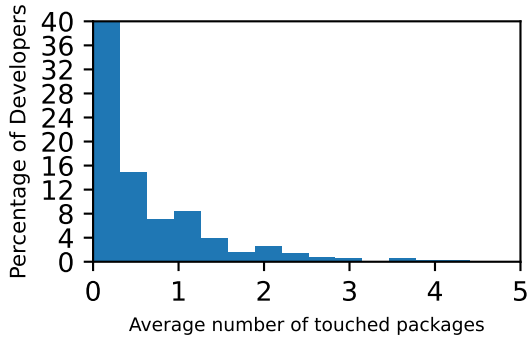


(f) Sustained later joiner (n=203).

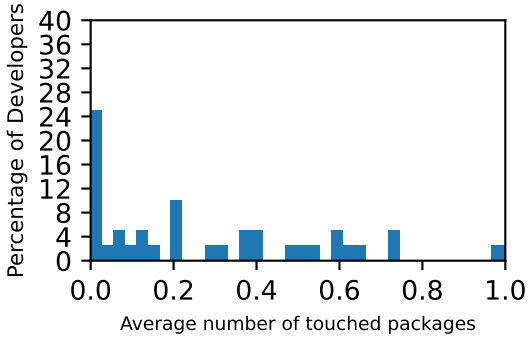
Figure 5: Histogram of the number of developers (y-axis) against the average packages touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



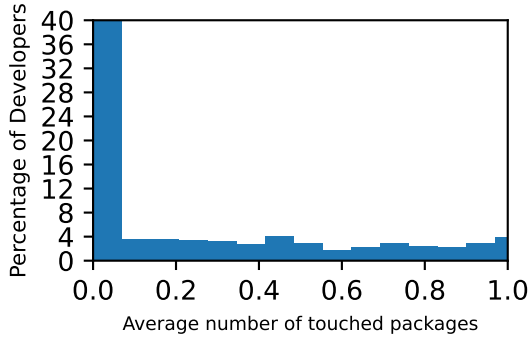
(a) Transient founder (n=28).



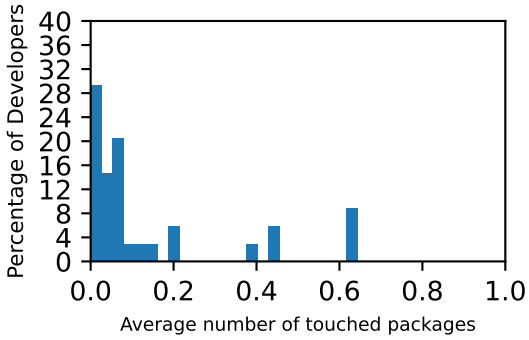
(b) Transient later joiner (n=1396).



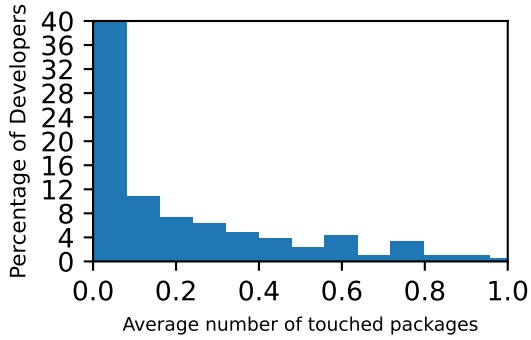
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

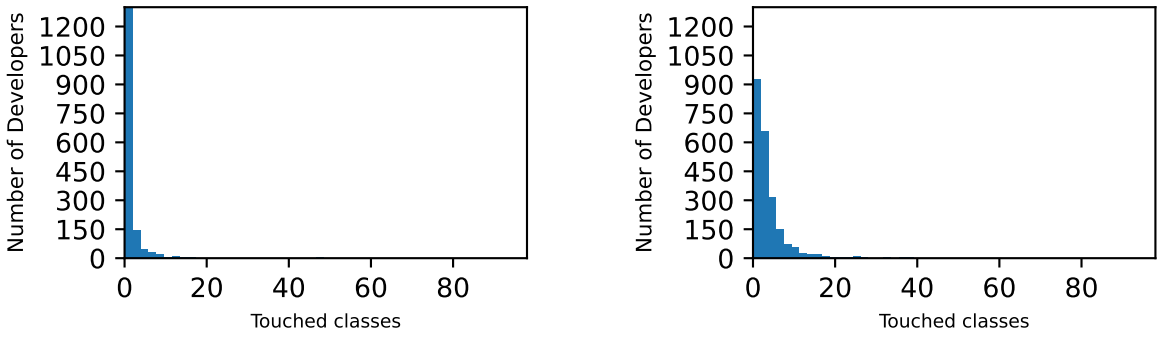


(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

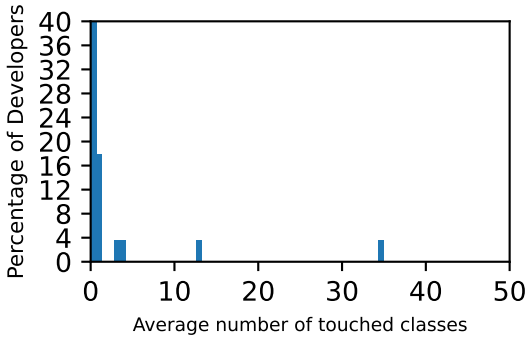
Figure 6: Histogram of the number of developers (y-axis) against the average packages touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



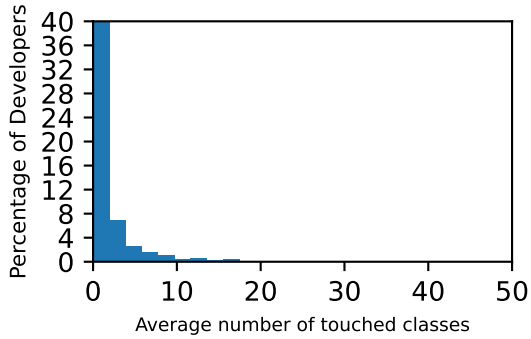
(a) By Day.

(b) By Commit.

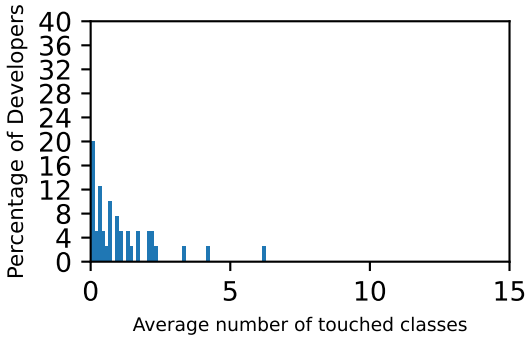
Figure 7: Histogram of the number of developers (y-axis) against the average classes touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



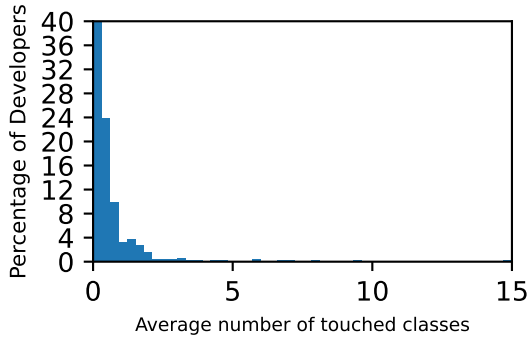
(a) Transient founder (n=28).



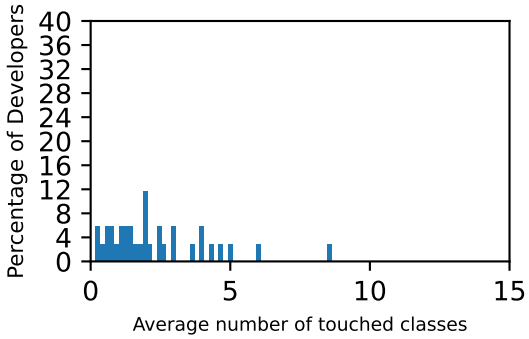
(b) Transient later joiner (n=1396).



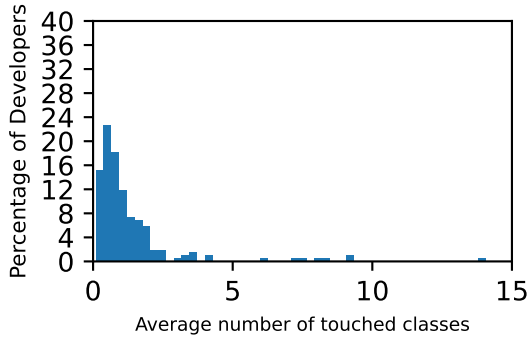
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

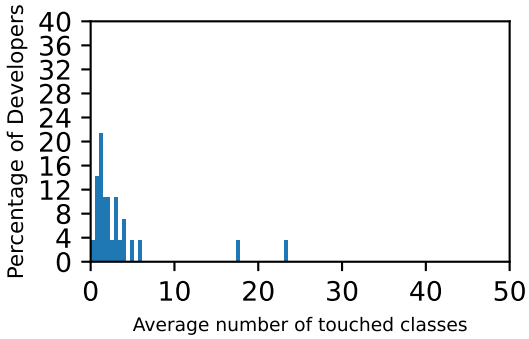


(e) Sustained founder (n=34).

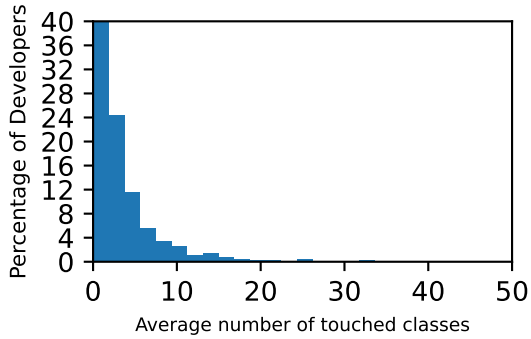


(f) Sustained later joiner (n=203).

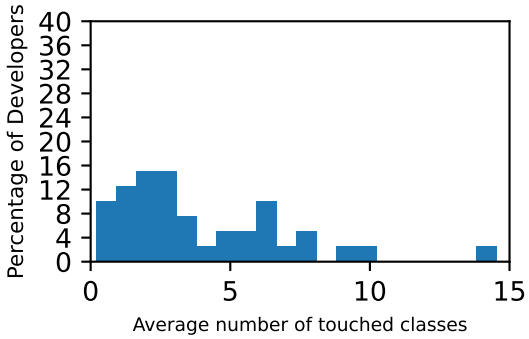
Figure 8: Histogram of the number of developers (y-axis) against the average classes touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



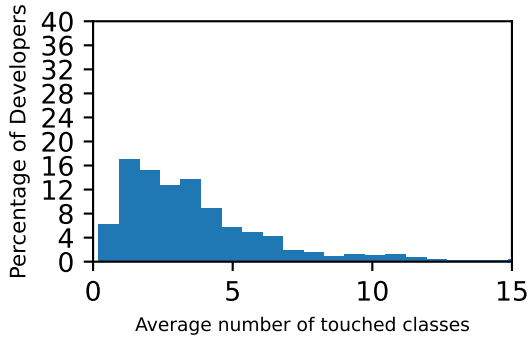
(a) Transient founder (n=28).



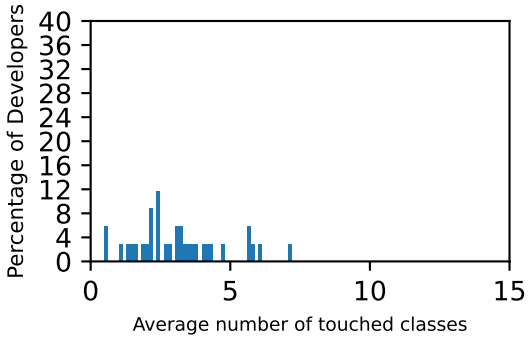
(b) Transient later joiner (n=1396).



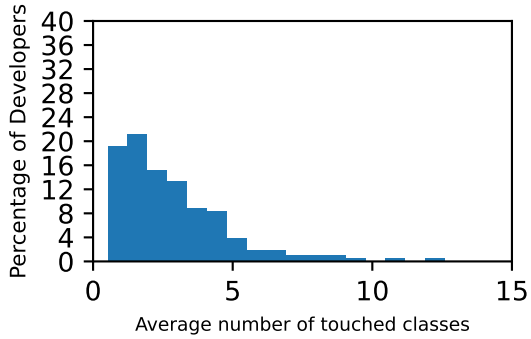
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

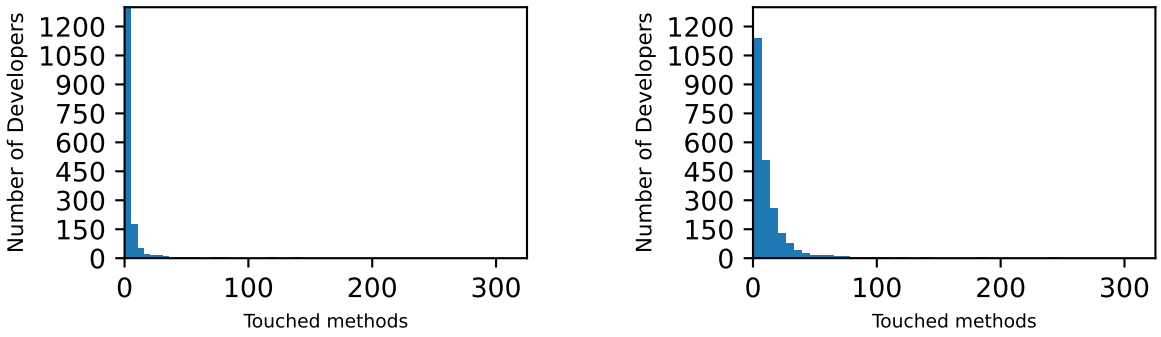


(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

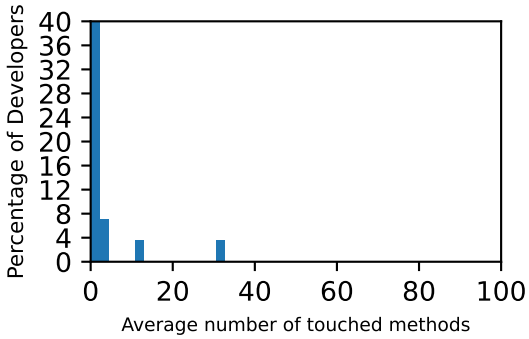
Figure 9: Histogram of the number of developers (y-axis) against the average classes touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



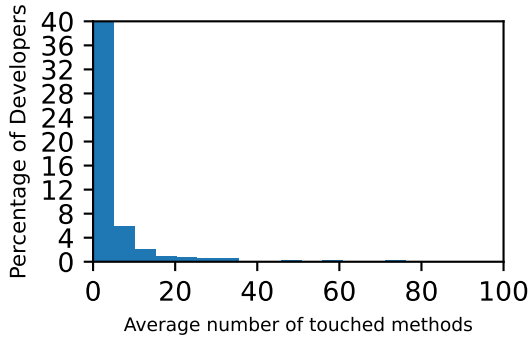
(a) By Day.

(b) By Commit.

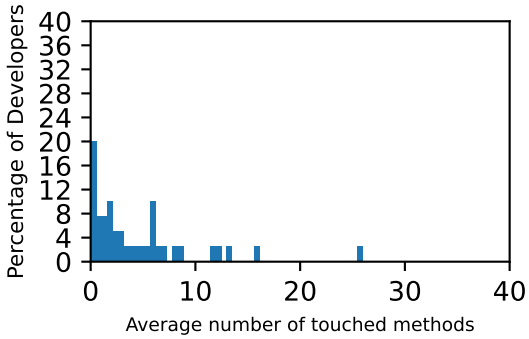
Figure 10: Histogram of the number of developers (y-axis) against the average methods touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



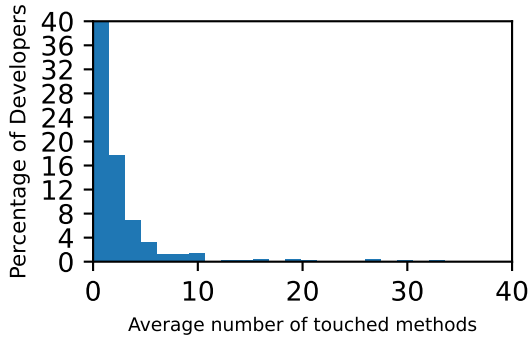
(a) Transient founder (n=28).



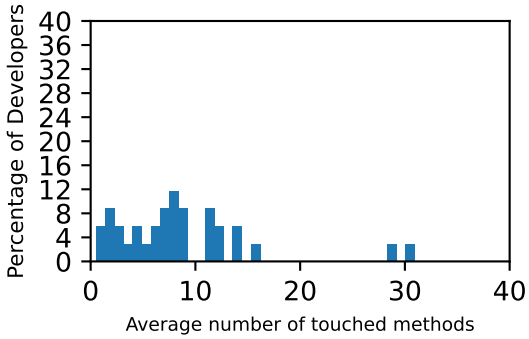
(b) Transient later joiner (n=1396).



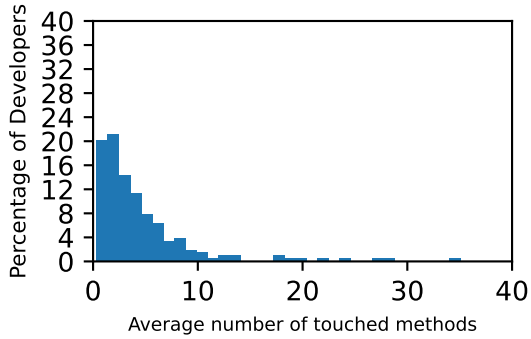
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).

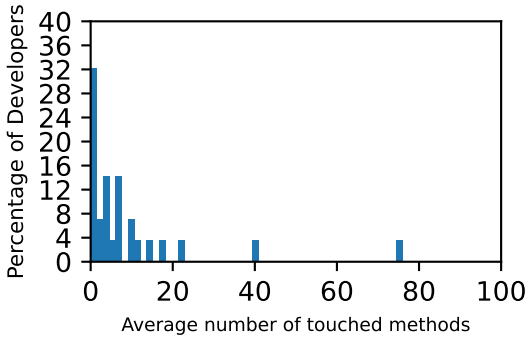


(e) Sustained founder (n=34).

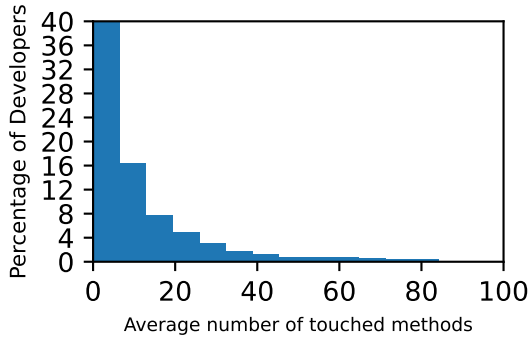


(f) Sustained later joiner (n=203).

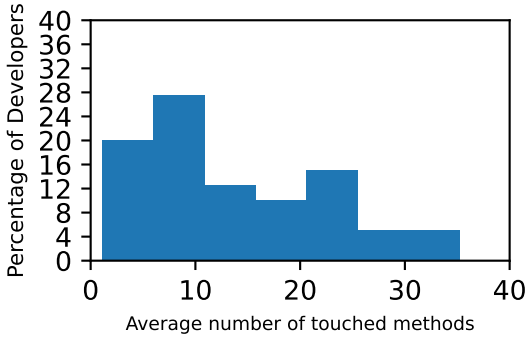
Figure 11: Histogram of the number of developers (y-axis) against the average methods touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



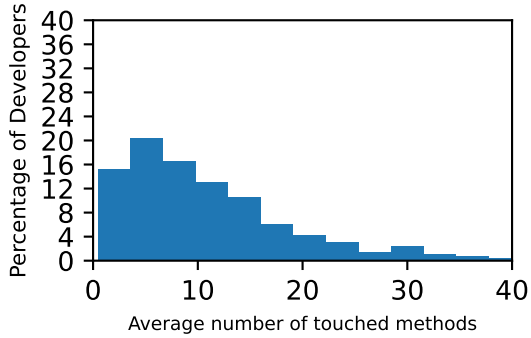
(a) Transient founder (n=28).



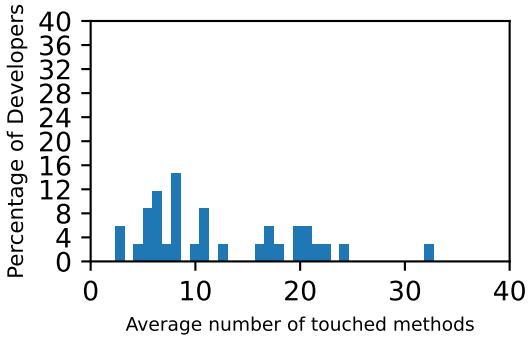
(b) Transient later joiner (n=1396).



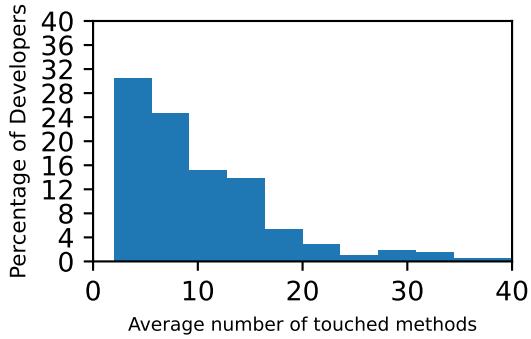
(c) Moderate founder (n=40).



(d) Moderate later joiner (n=580).



(e) Sustained founder (n=34).



(f) Sustained later joiner (n=203).

Figure 12: Histogram of the number of developers (y-axis) against the average methods touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.

1.6 Time series components touched - For all components touched for first five commits and for each period

1.6.1 Time series components touched - For packages touched for first five commits

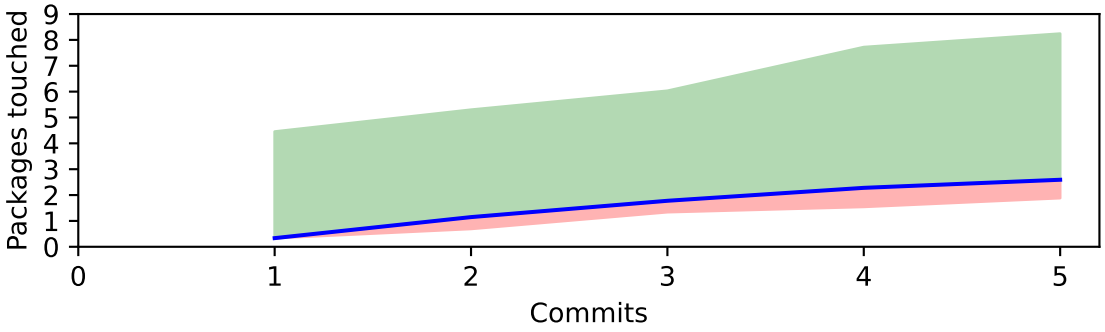


Figure 13: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

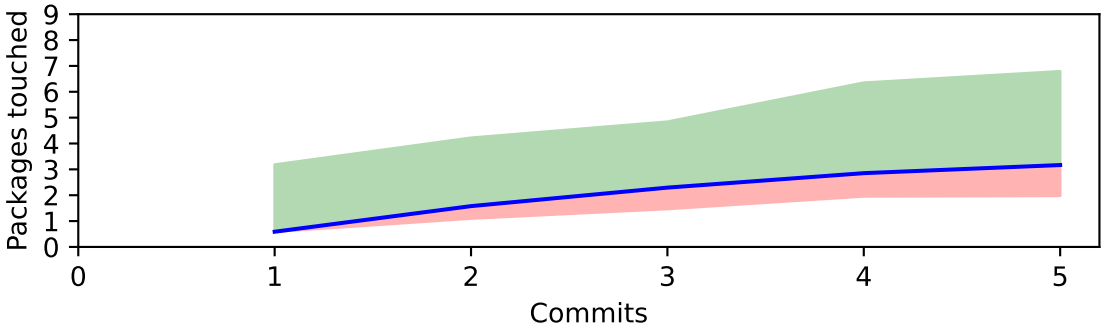


Figure 14: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

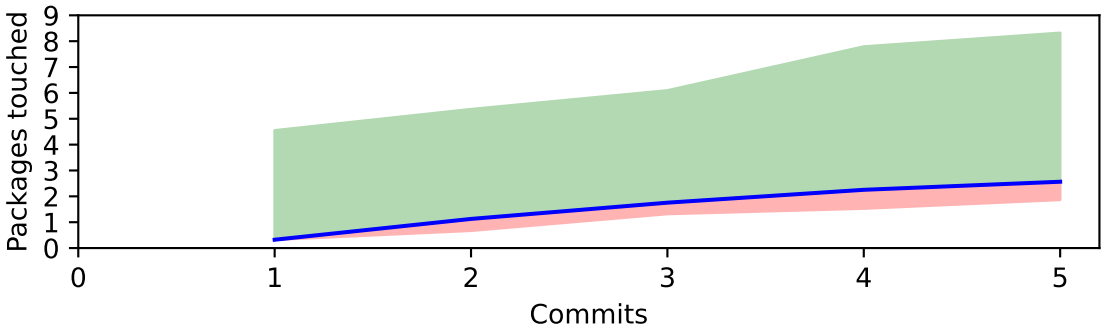


Figure 15: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

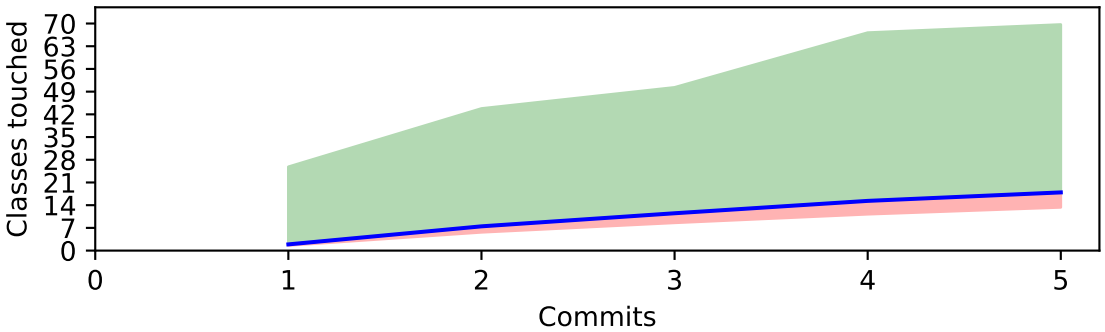


Figure 16: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

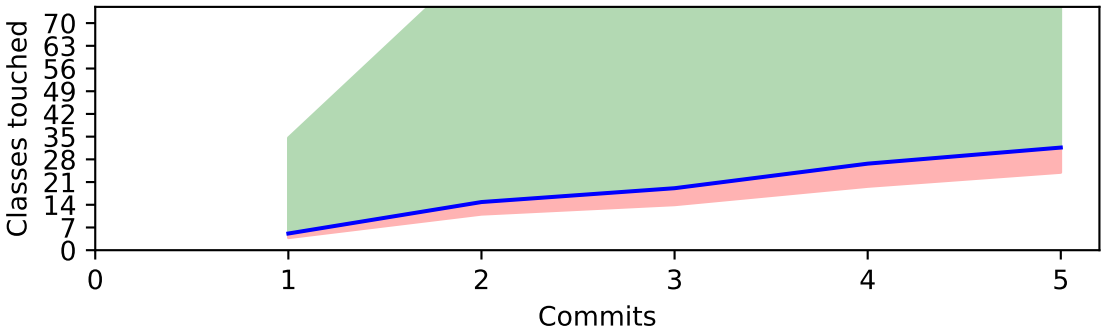


Figure 17: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

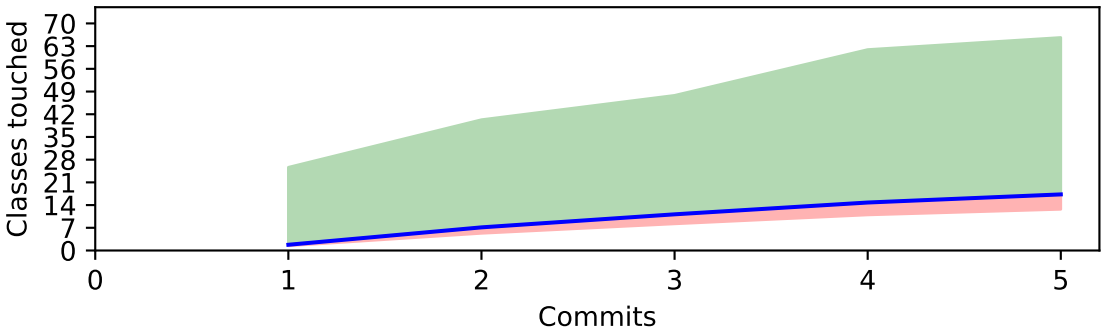


Figure 18: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

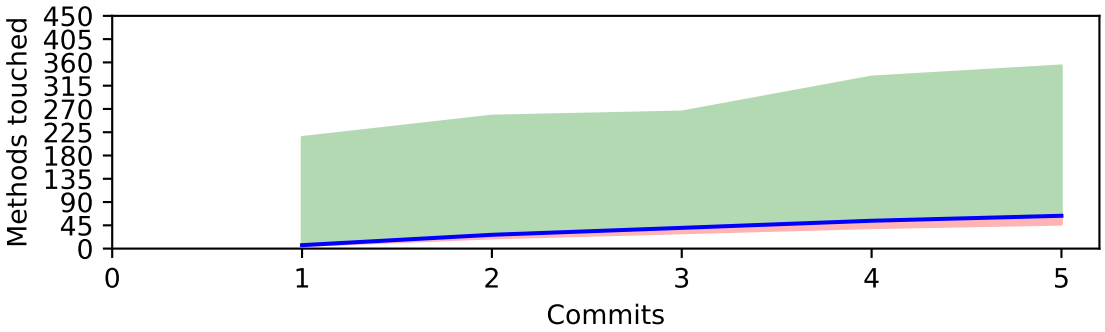


Figure 19: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

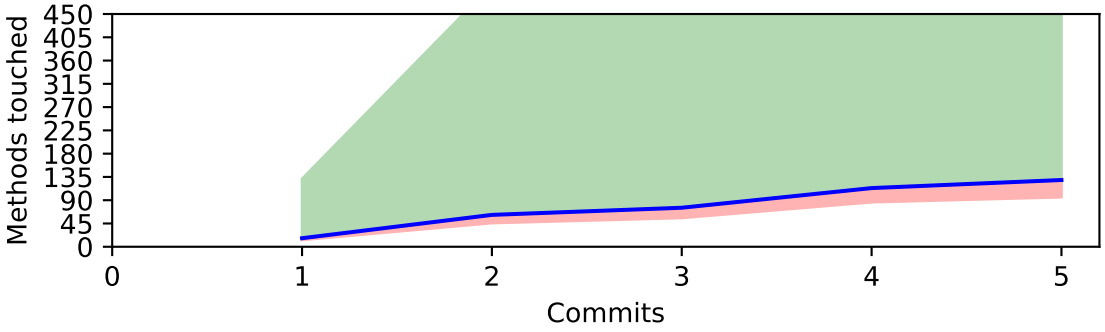


Figure 20: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

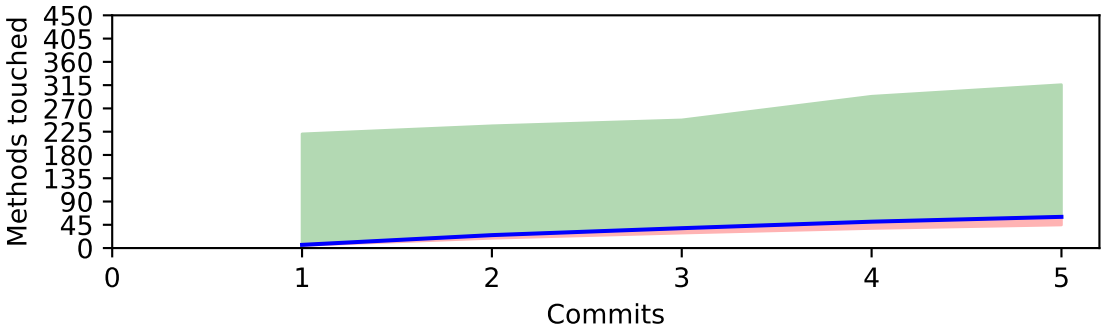
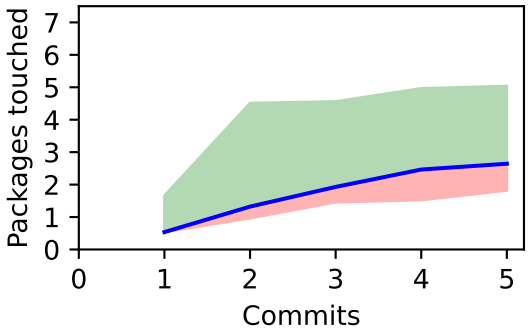


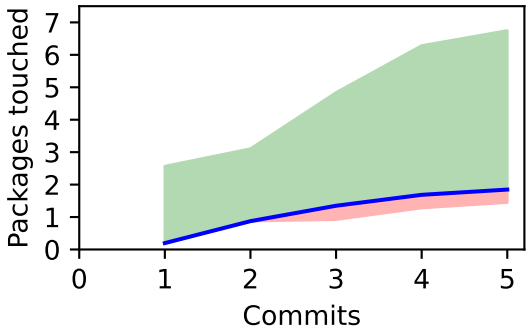
Figure 21: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

1.6.2 Time series components touched - For packages touched for each week first five commits

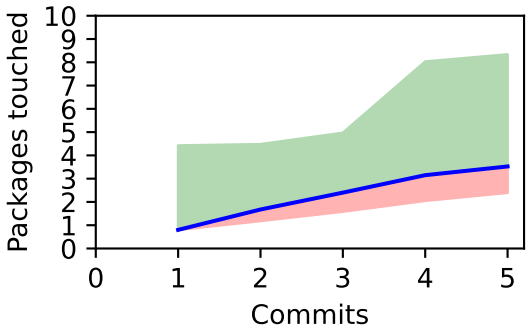
1.6.3 Time series components touched - For packages touched for first ten commits



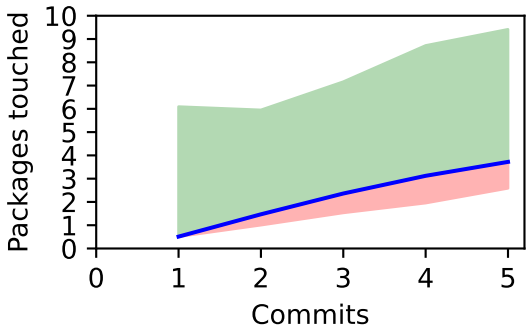
(a) Transient founder (n=28 & μ=2.64)



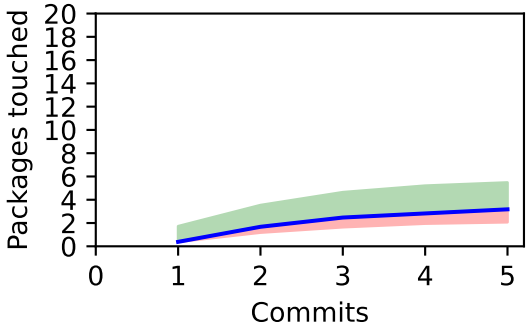
(b) Transient later joiner (n=1396 & μ=1.85)



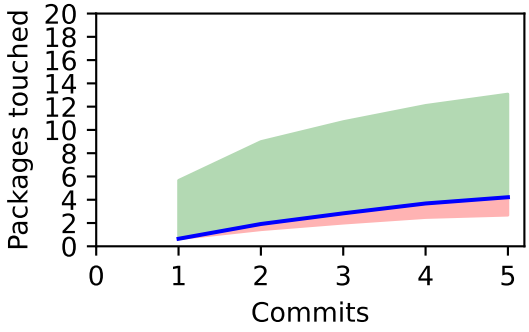
(c) Moderate founder (n=40 & μ=3.52)



(d) Moderate later joiner (n=580 & μ=3.72)

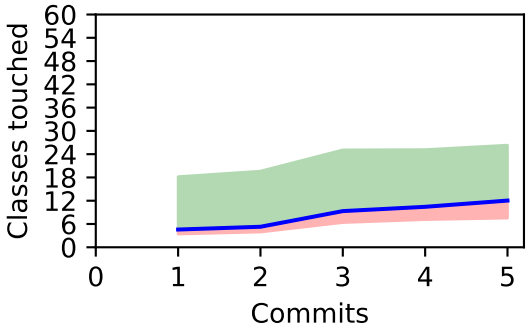


(e) Sustained founder (n=34 & μ=3.18)

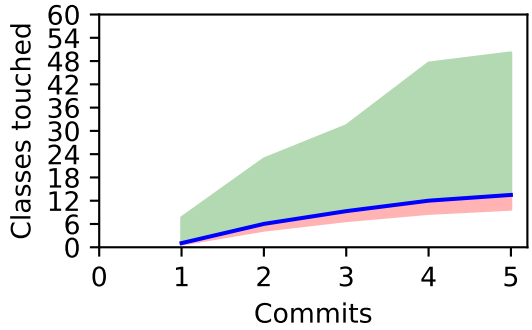


(f) Sustained later joiner (n=203 & μ=4.22)

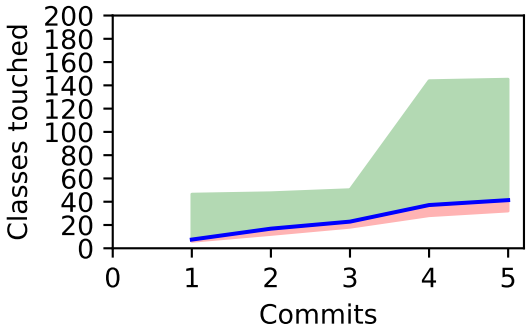
Figure 22: A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



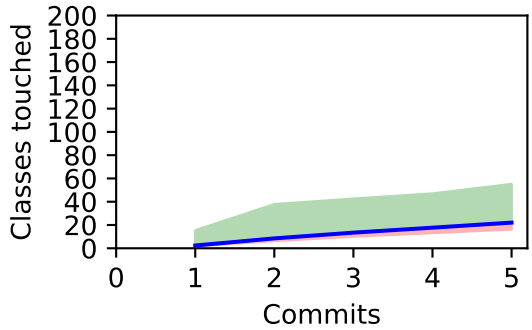
(a) Transient founder (n=28 & μ=12.04)



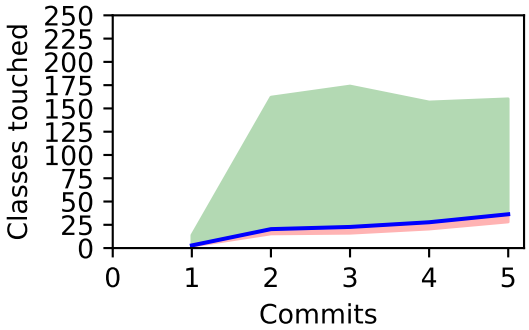
(b) Transient later joiner (n=1396 & μ=13.49)



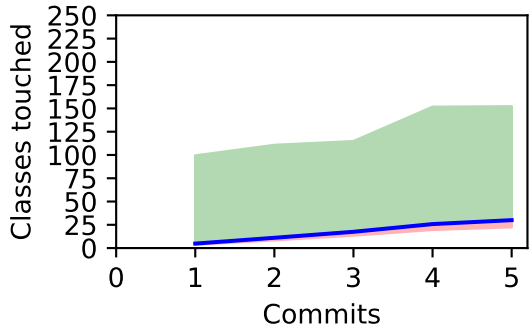
(c) Moderate founder (n=40 & μ=41.38)



(d) Moderate later joiner (n=580 & μ=22.04)

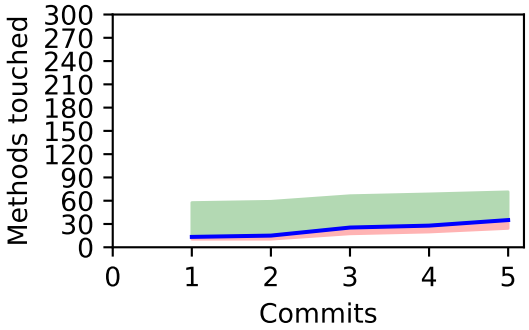


(e) Sustained founder (n=34 & μ=36.32)

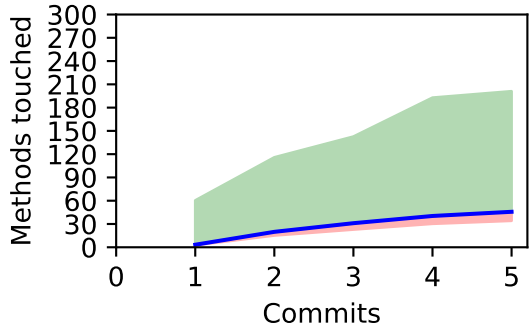


(f) Sustained later joiner (n=203 & μ=30.01)

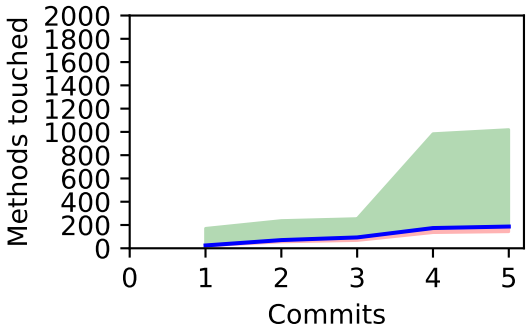
Figure 23: A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



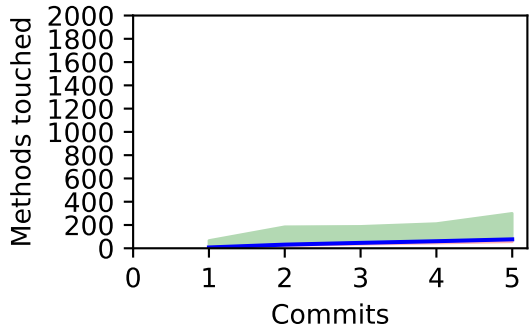
(a) Transient founder (n=28 & μ=35.07)



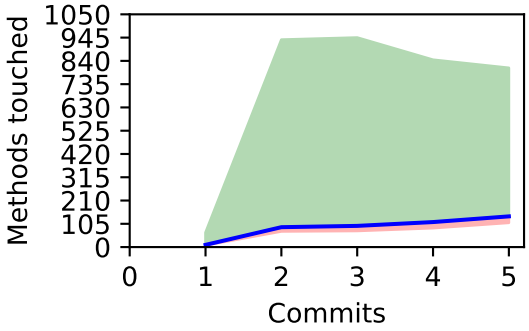
(b) Transient later joiner (n=1396 & μ=45.72)



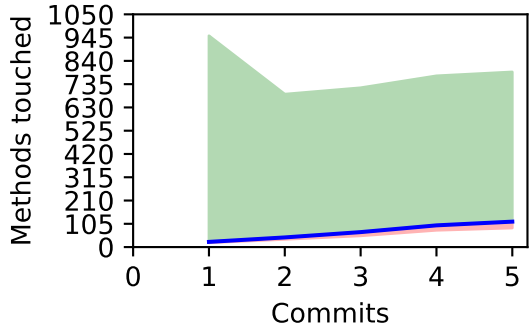
(c) Moderate founder (n=40 & μ=186.75)



(d) Moderate later joiner (n=580 & μ=76.85)



(e) Sustained founder (n=34 & μ=138.53)



(f) Sustained later joiner (n=203 & μ=114.77)

Figure 24: A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

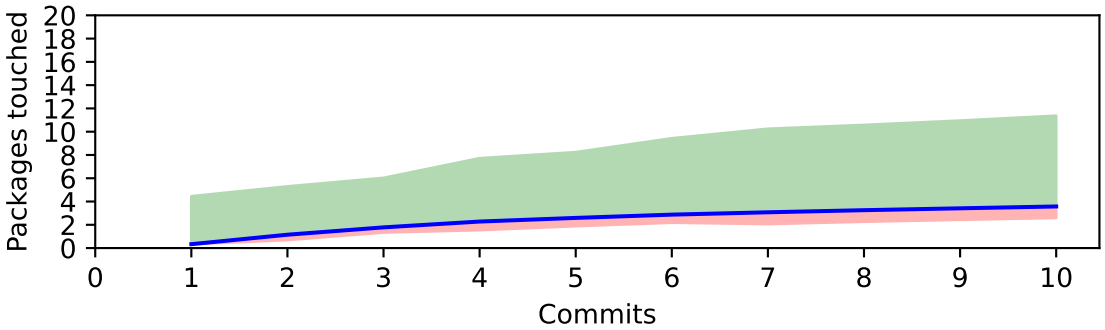


Figure 25: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

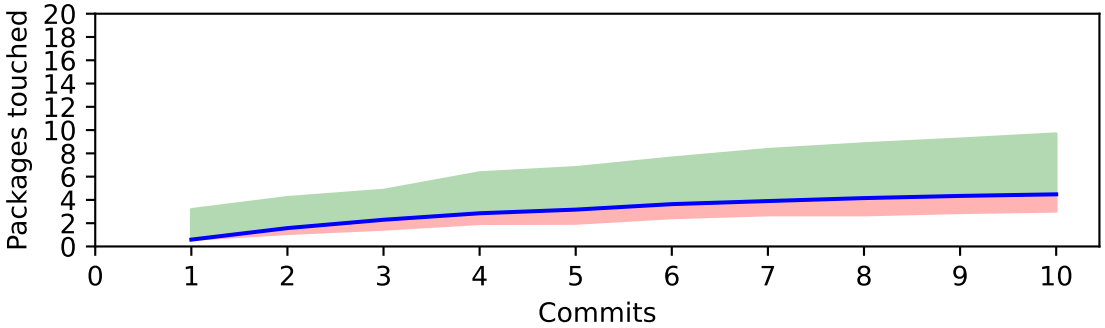


Figure 26: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

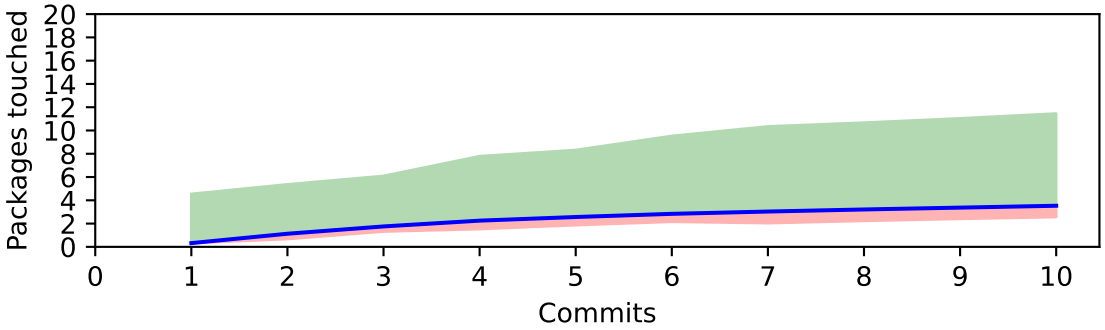


Figure 27: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

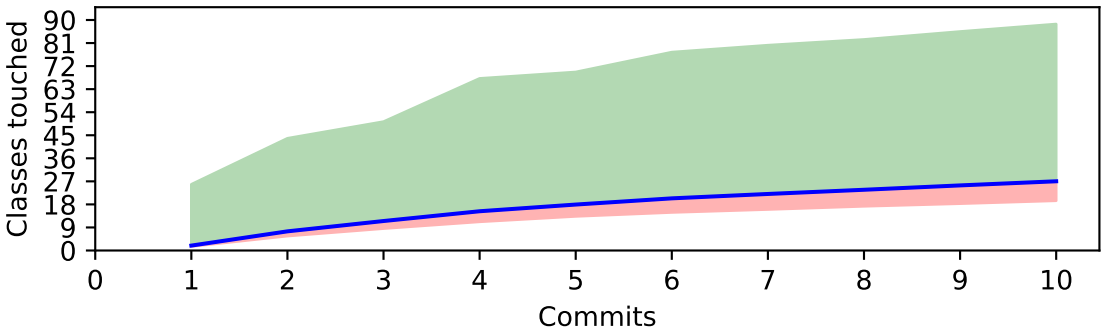


Figure 28: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

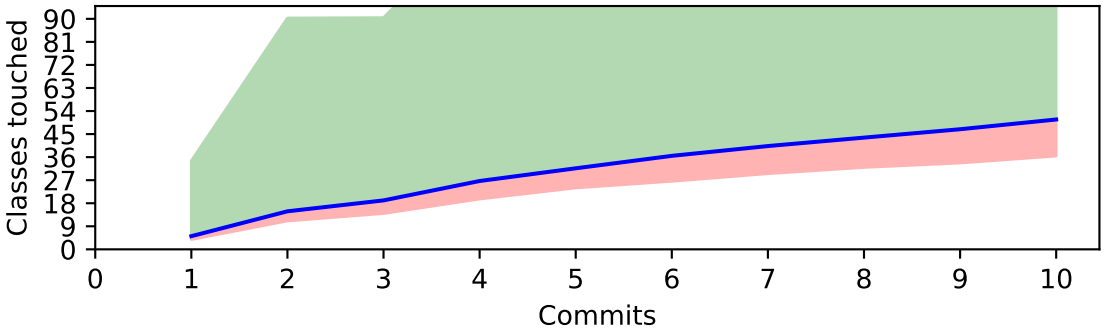


Figure 29: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

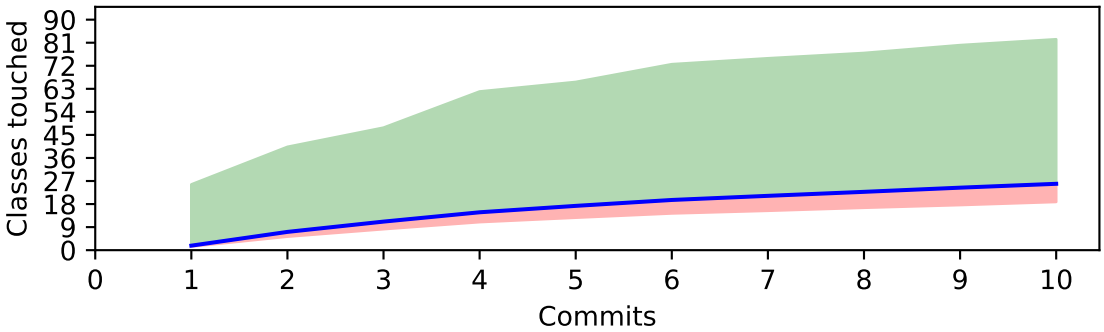


Figure 30: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

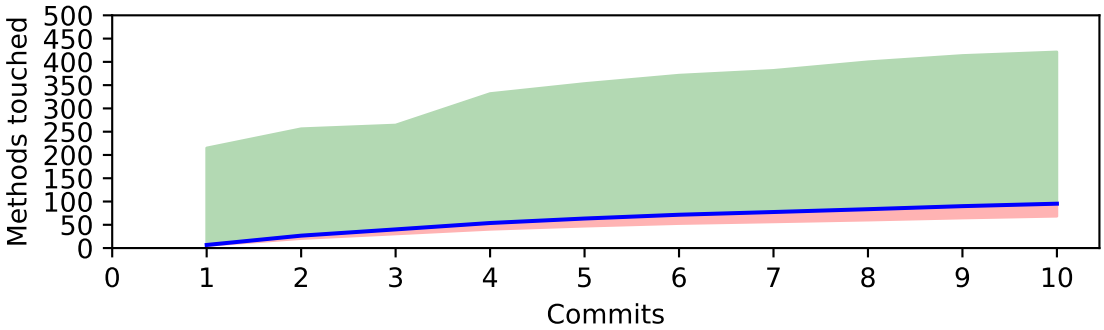


Figure 31: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

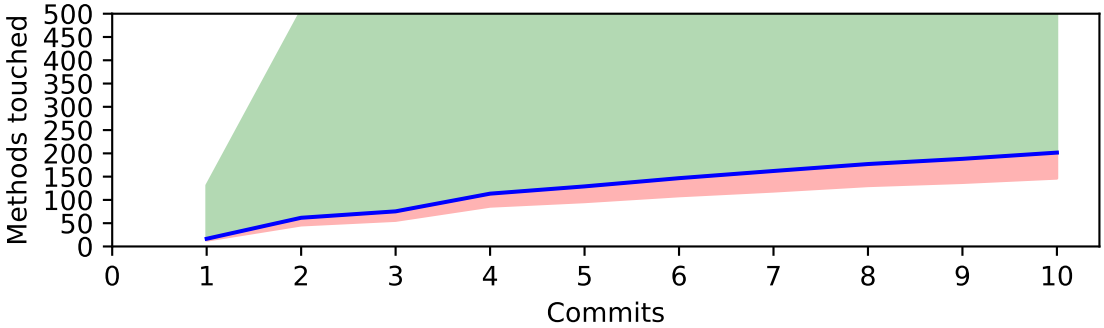


Figure 32: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

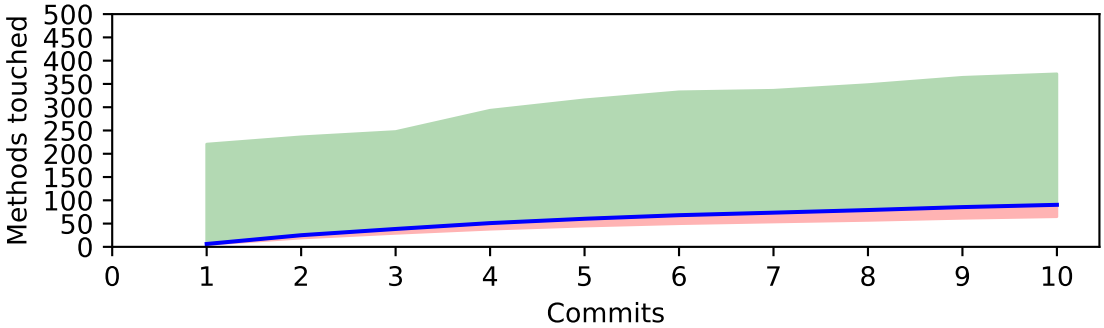
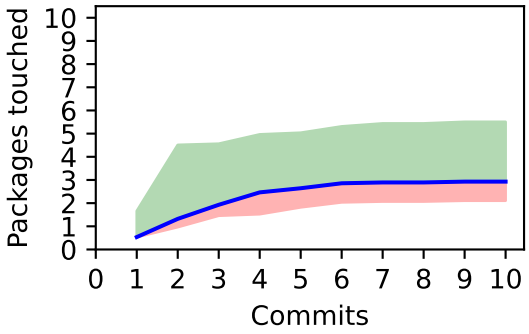


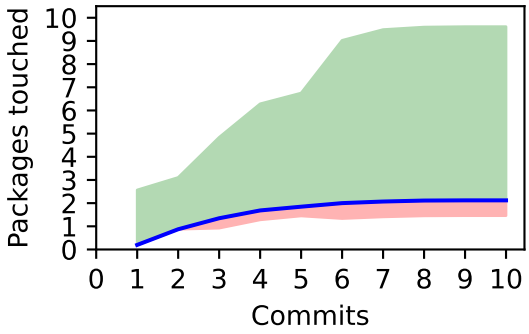
Figure 33: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

1.6.4 Time series components touched - For packages touched for each week first ten commits

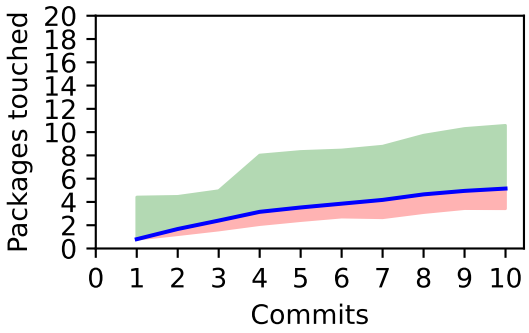
1.6.5 Time series components touched - For packages touched for first twenty commits



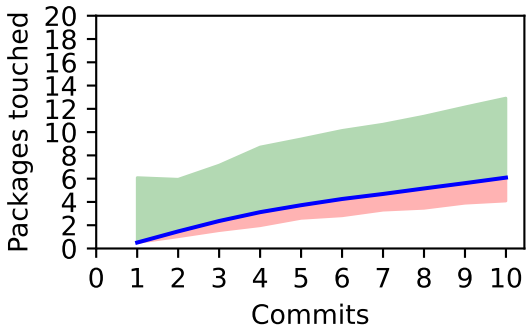
(a) Transient founder (n=28 & μ=2.93)



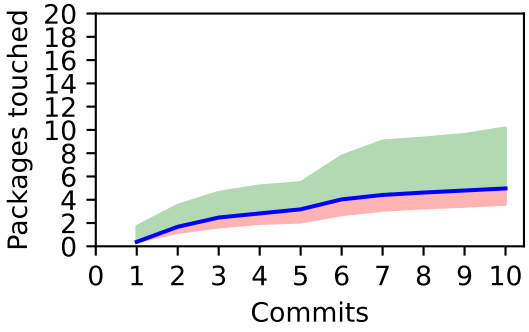
(b) Transient later joiner (n=1396 & μ=2.12)



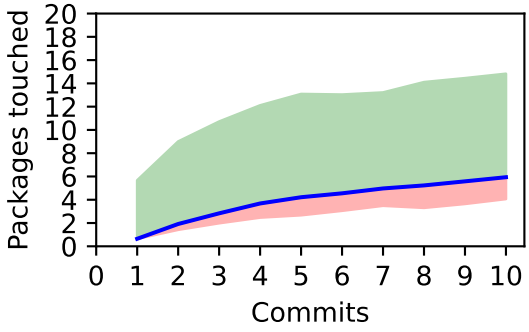
(c) Moderate founder (n=40 & μ=5.15)



(d) Moderate later joiner (n=580 & μ=6.09)

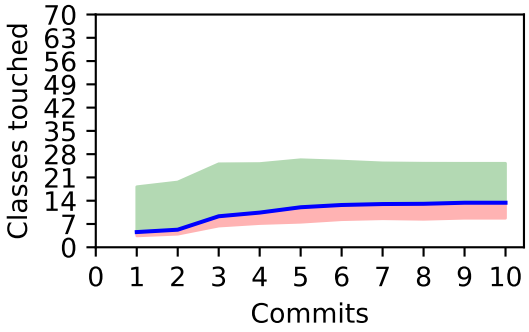


(e) Sustained founder (n=34 & μ=4.97)

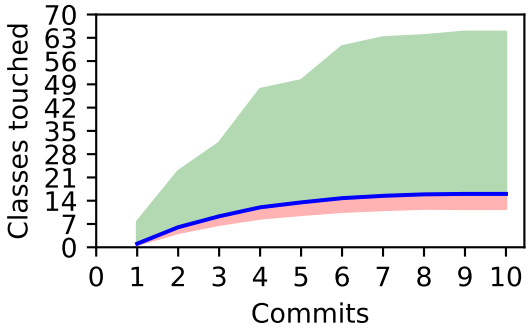


(f) Sustained later joiner (n=203 & μ=5.94)

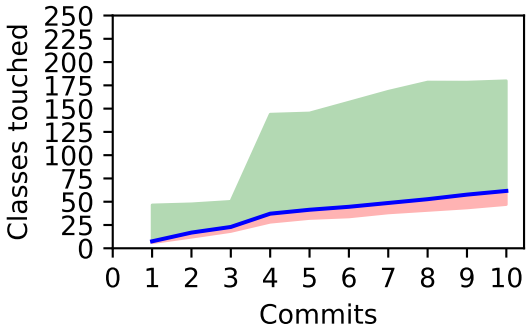
Figure 34: A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



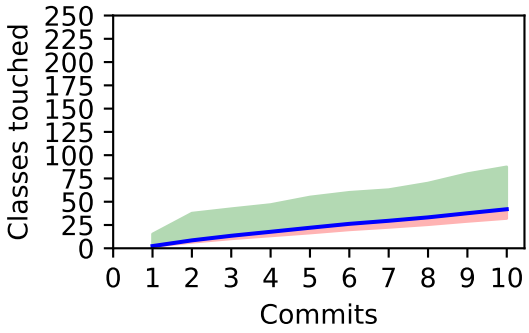
(a) Transient founder (n=28 & μ=13.39)



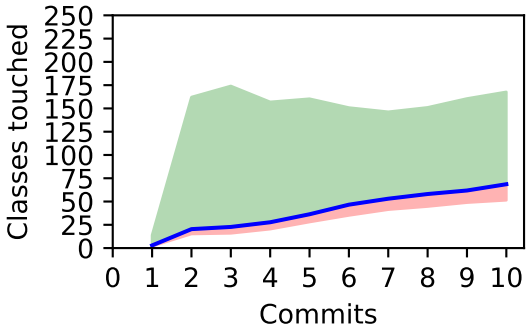
(b) Transient later joiner (n=1396 & μ=16.05)



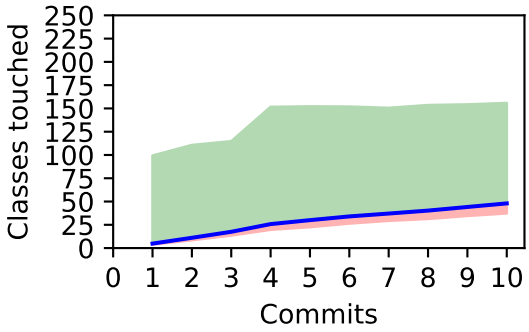
(c) Moderate founder (n=40 & μ=61.6)



(d) Moderate later joiner (n=580 & μ=41.95)

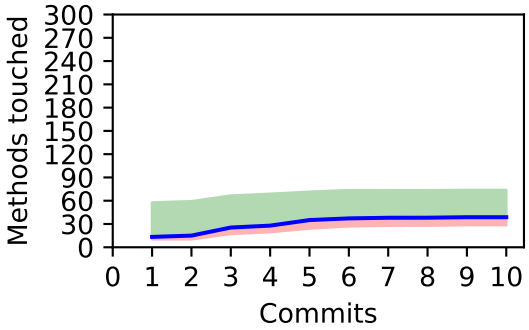


(e) Sustained founder (n=34 & μ=68.59)

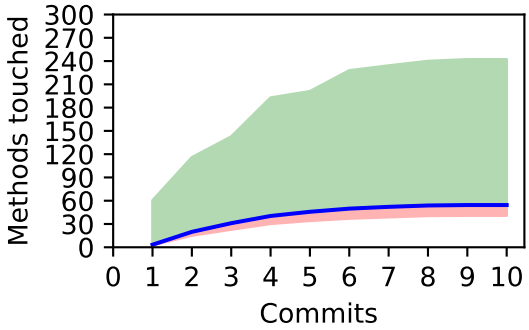


(f) Sustained later joiner (n=203 & μ=47.98)

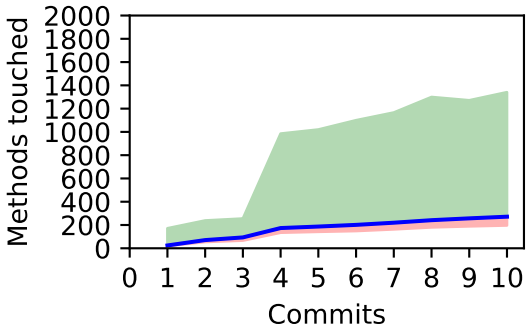
Figure 35: A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



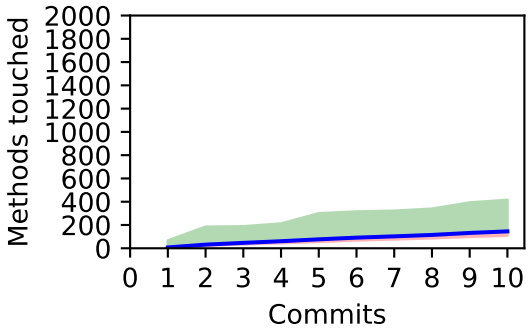
(a) Transient founder (n=28 & μ=38.75)



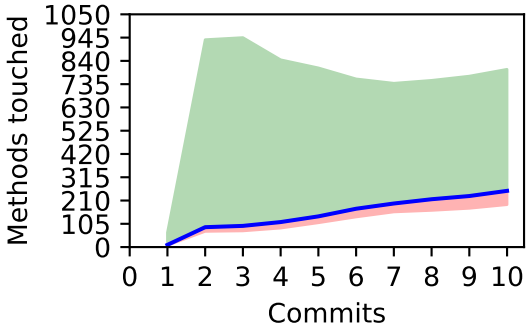
(b) Transient later joiner (n=1396 & μ=54.43)



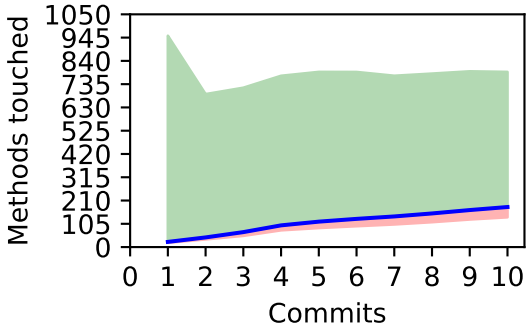
(c) Moderate founder (n=40 & μ=271.42)



(d) Moderate later joiner (n=580 & μ=145.21)



(e) Sustained founder (n=34 & μ=253.65)



(f) Sustained later joiner (n=203 & μ=180.7)

Figure 36: A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

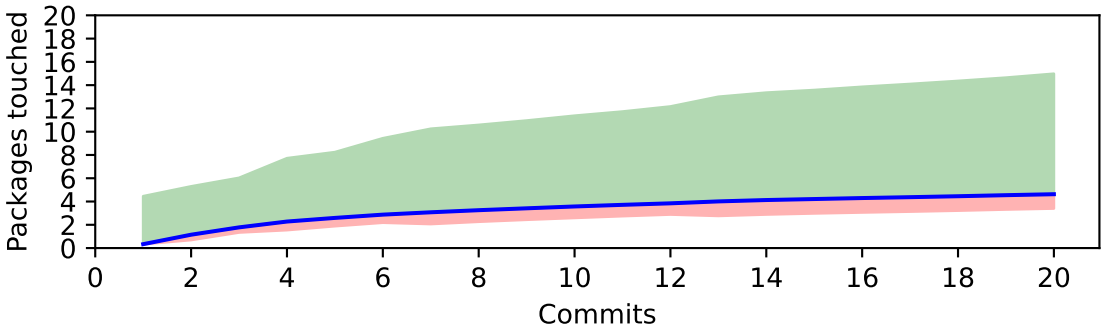


Figure 37: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

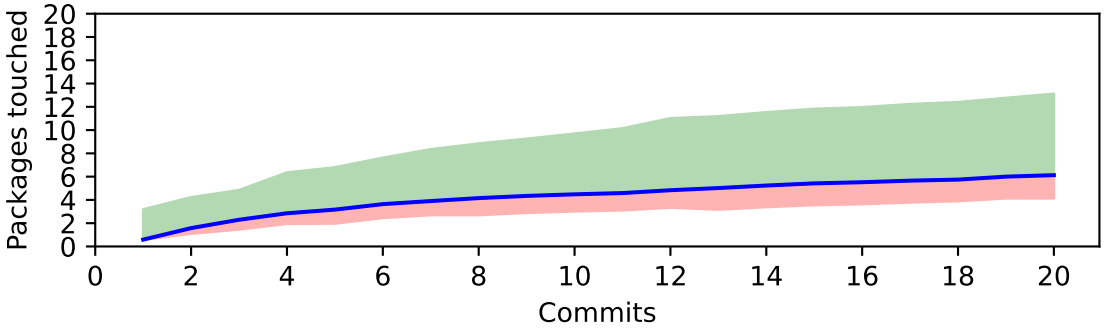


Figure 38: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

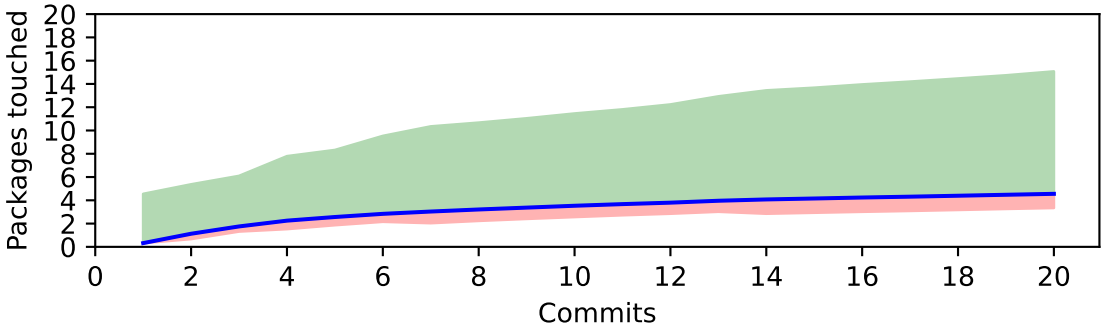


Figure 39: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

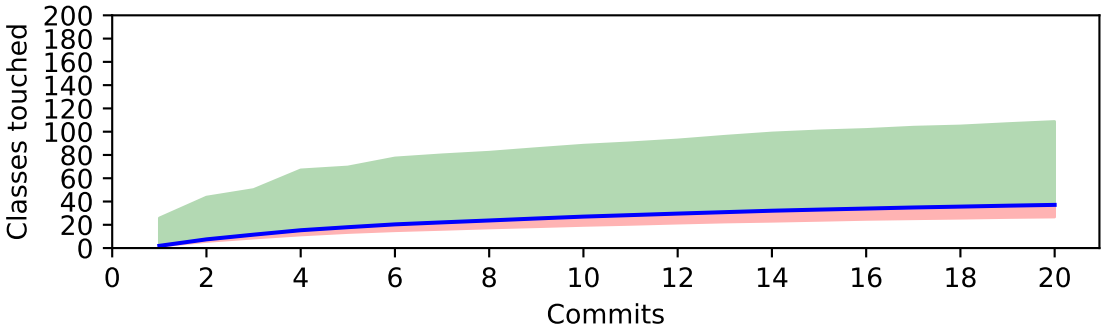


Figure 40: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

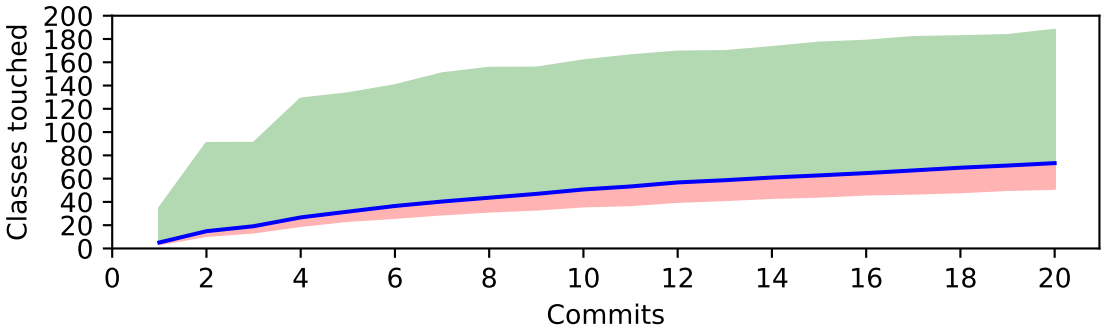


Figure 41: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

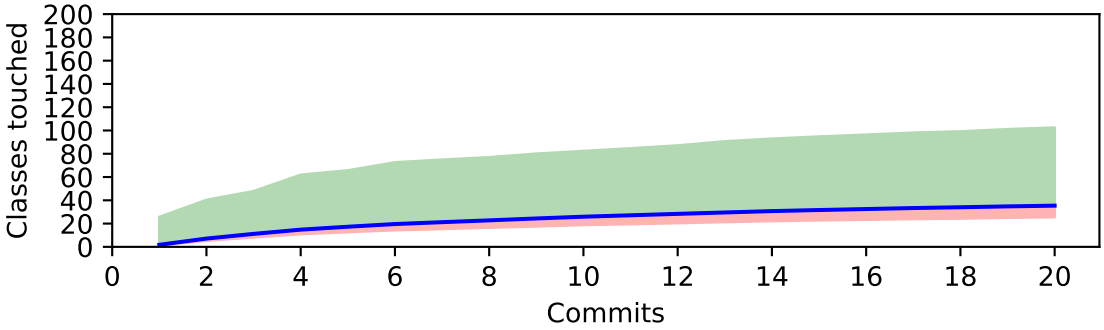


Figure 42: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

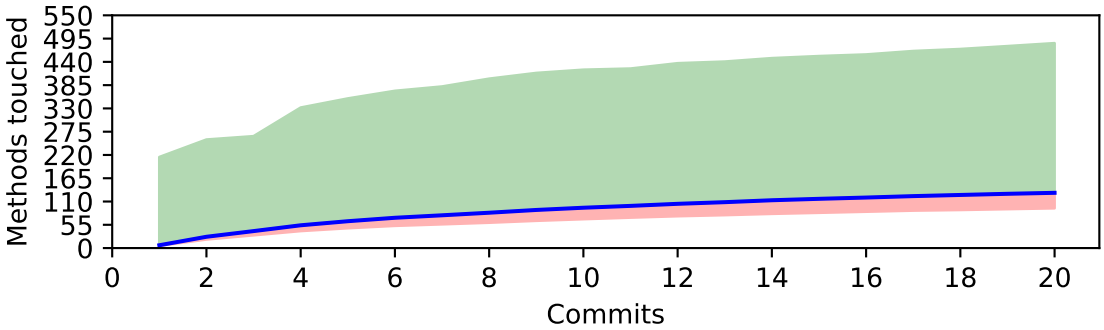


Figure 43: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

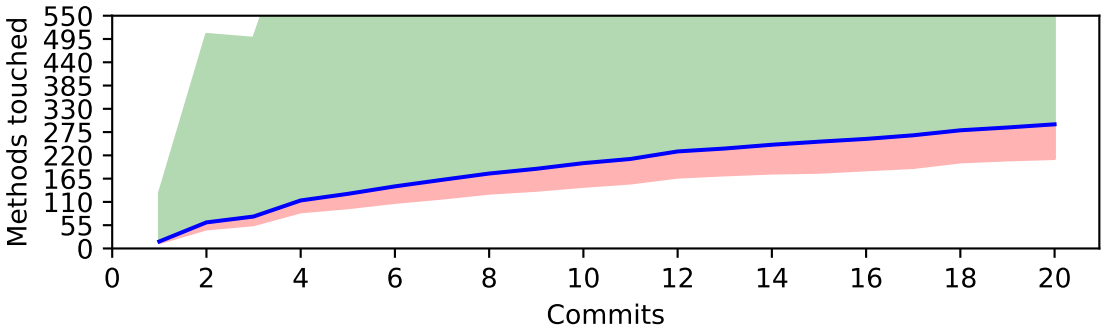


Figure 44: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

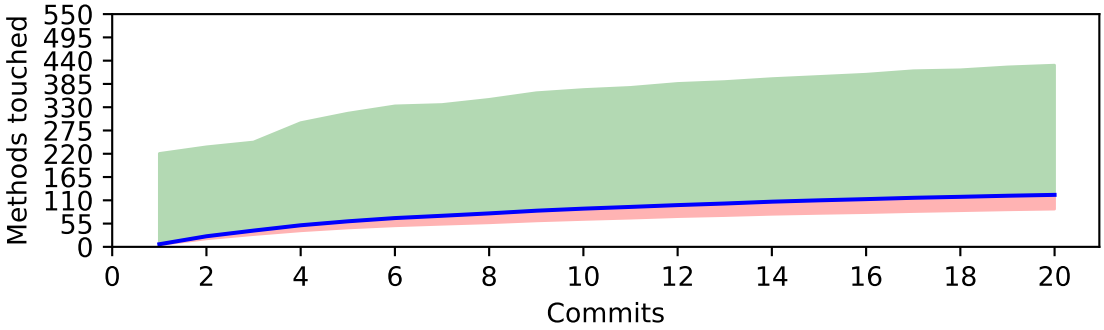
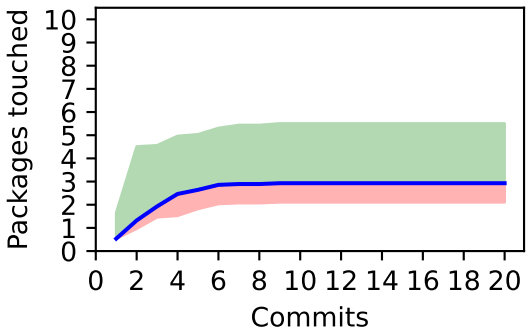


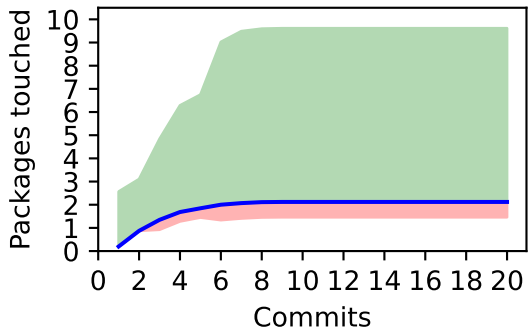
Figure 45: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

1.6.6 Time series components touched - For packages touched for each week first twenty commits

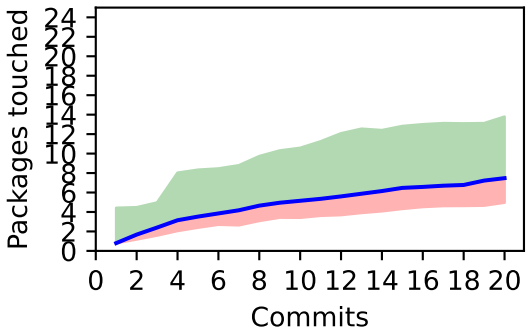
1.6.7 Time series components touched - For packages touched for commit



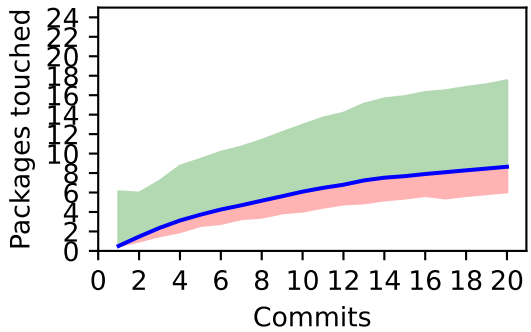
(a) Transient founder (n=28 & μ=2.93)



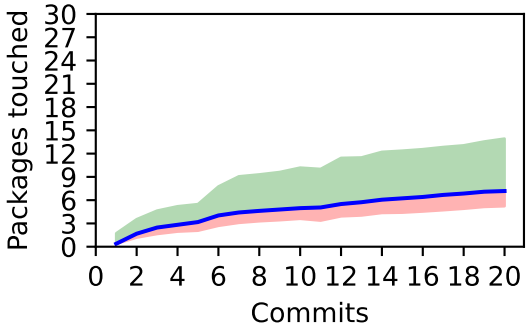
(b) Transient later joiner (n=1396 & μ=2.12)



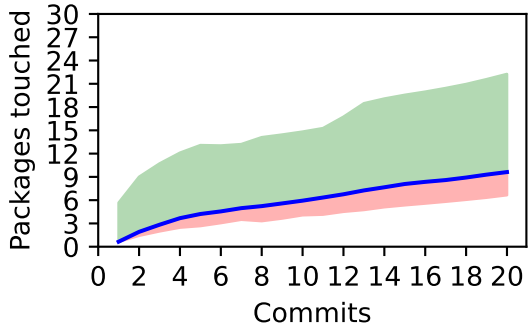
(c) Moderate founder (n=40 & μ=7.48)



(d) Moderate later joiner (n=580 & μ=8.65)

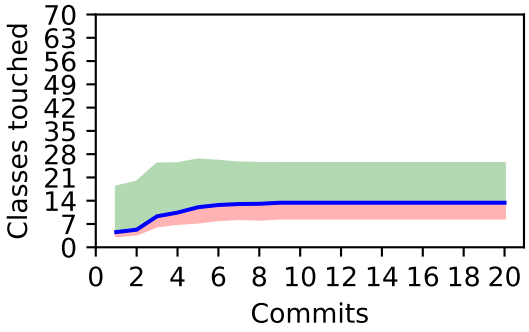


(e) Sustained founder (n=34 & μ=7.18)

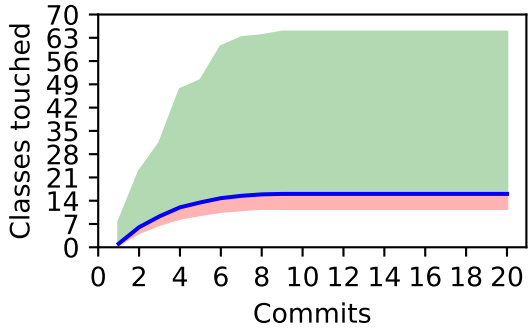


(f) Sustained later joiner (n=203 & μ=9.62)

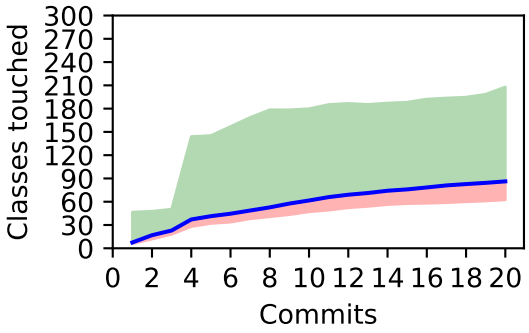
Figure 46: A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



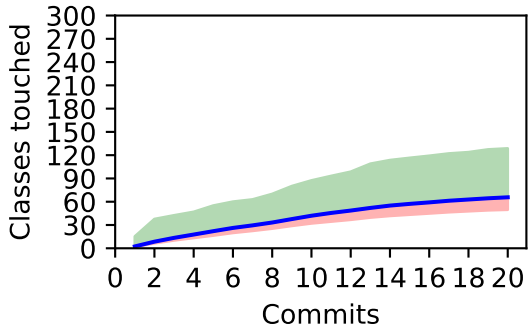
(a) Transient founder (n=28 & μ=13.39)



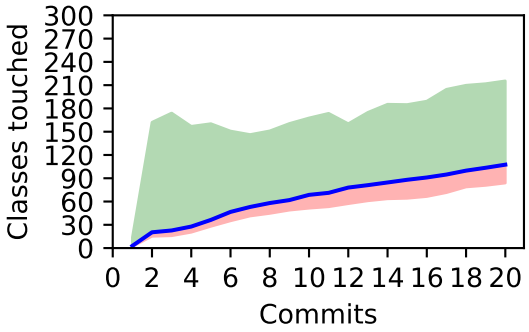
(b) Transient later joiner (n=1396 & μ=16.05)



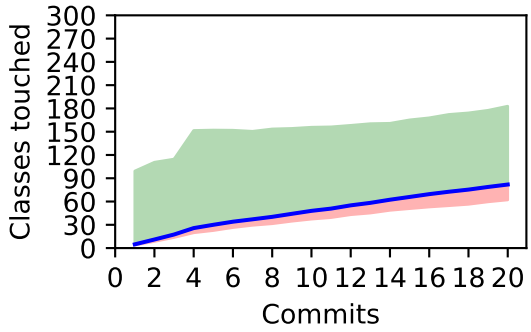
(c) Moderate founder (n=40 & μ=86.22)



(d) Moderate later joiner (n=580 & μ=65.7)

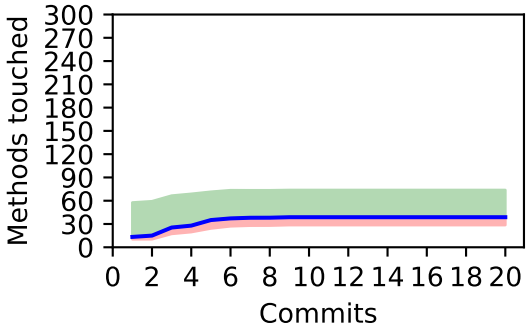


(e) Sustained founder (n=34 & μ=107.5)

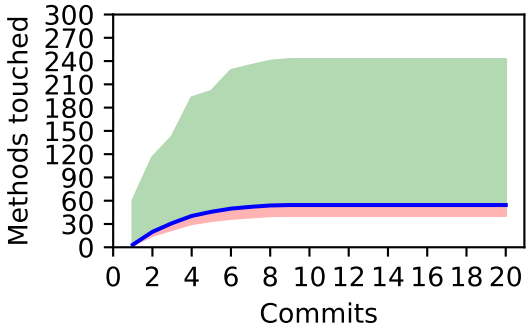


(f) Sustained later joiner (n=203 & μ=81.91)

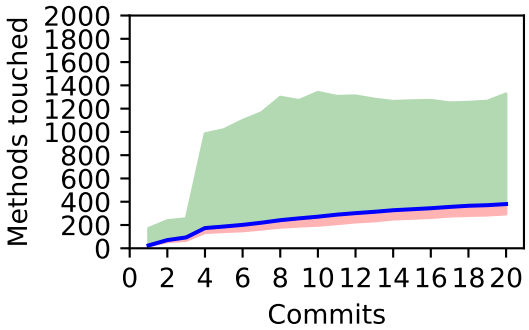
Figure 47: A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



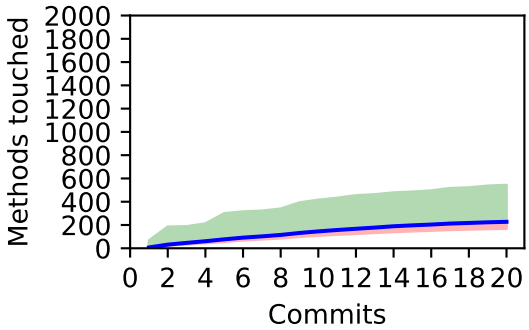
(a) Transient founder (n=28 & μ=38.75)



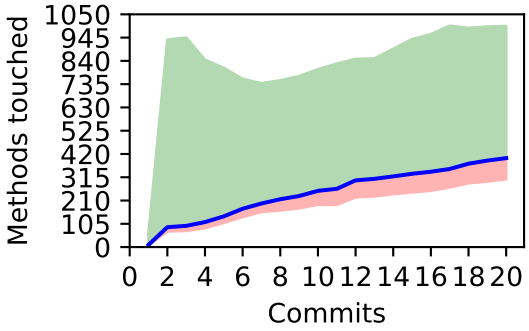
(b) Transient later joiner (n=1396 & μ=54.43)



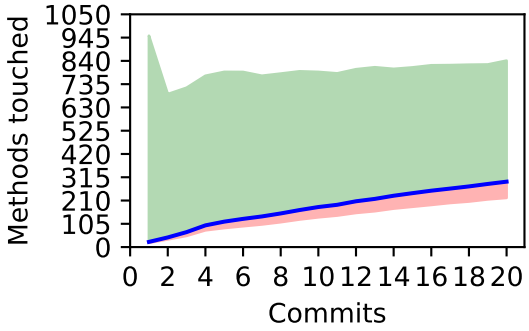
(c) Moderate founder (n=40 & μ=379.9)



(d) Moderate later joiner (n=580 & μ=227.38)



(e) Sustained founder (n=34 & μ=401.24)



(f) Sustained later joiner (n=203 & μ=295.1)

Figure 48: A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

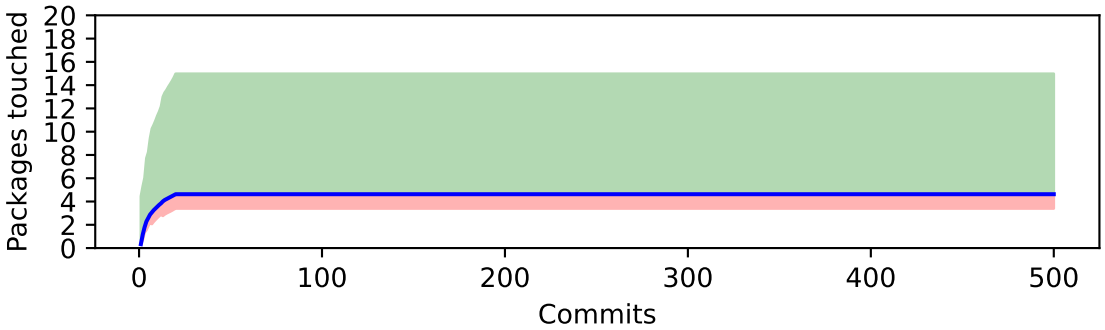


Figure 49: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

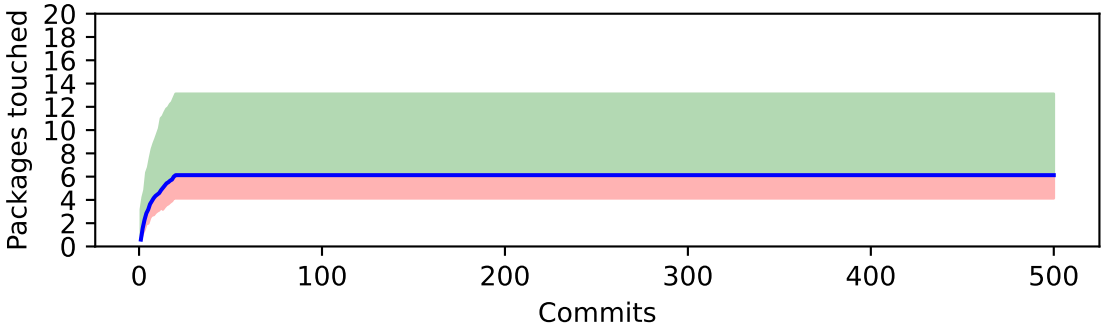


Figure 50: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

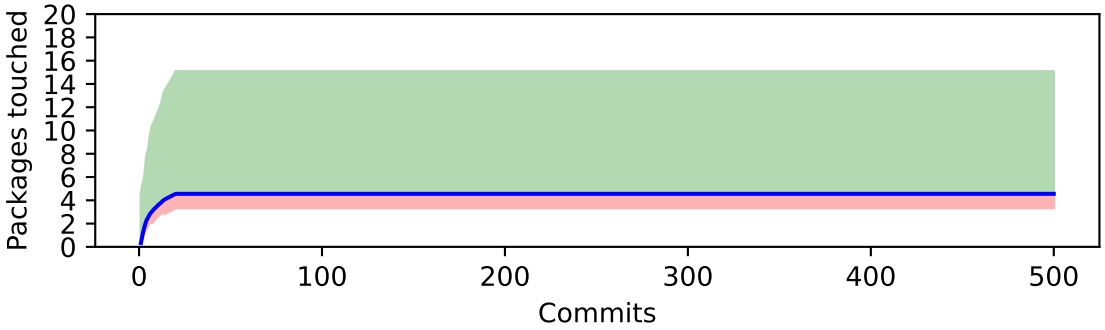


Figure 51: A time series of the average (mean) total packages touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

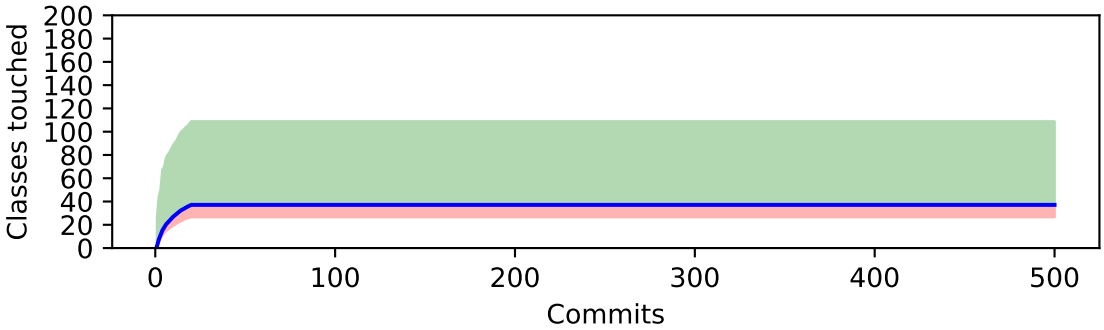


Figure 52: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

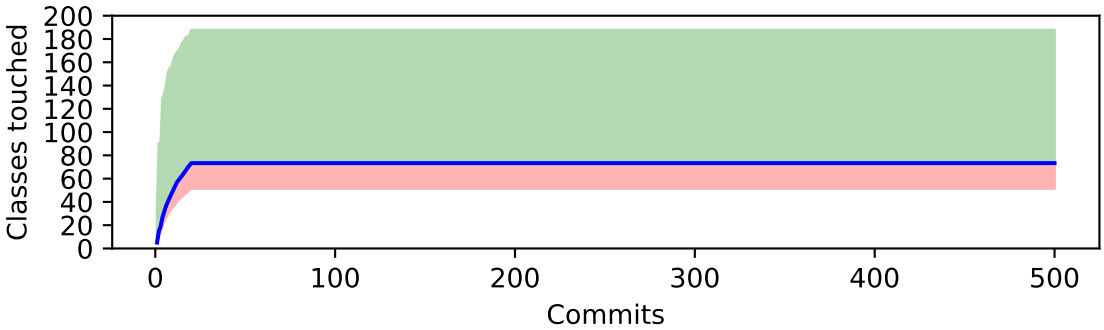


Figure 53: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

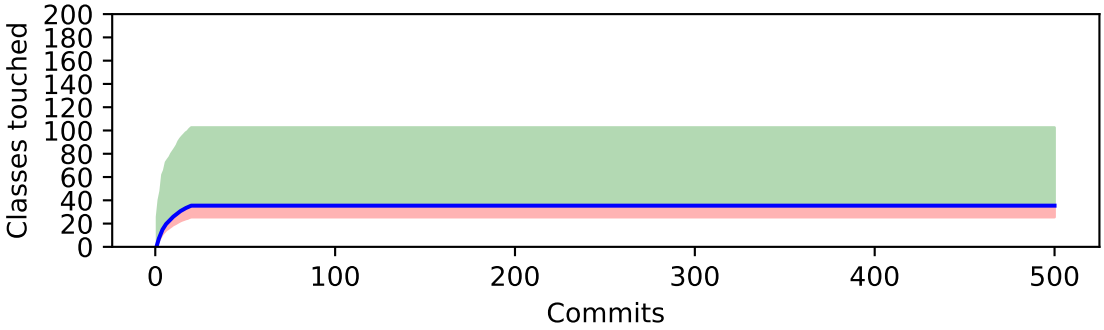


Figure 54: A time series of the average (mean) total classes touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

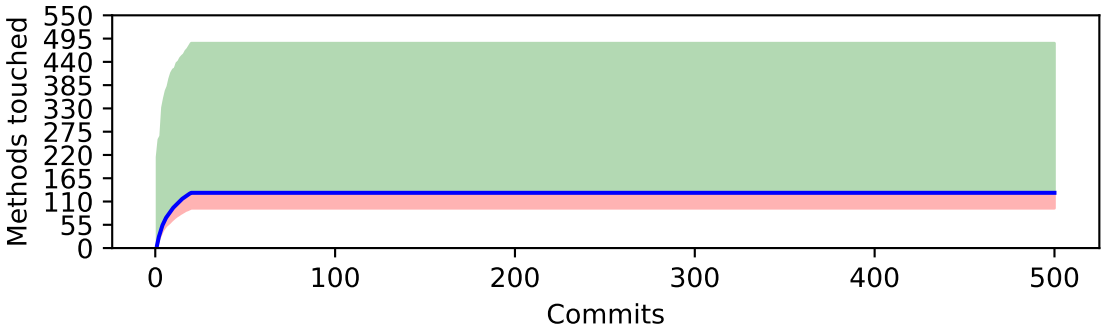


Figure 55: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All developers (n=2278).

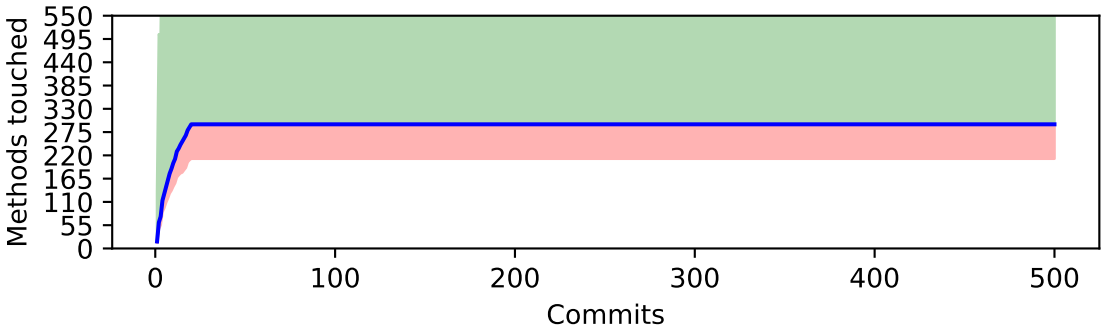


Figure 56: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Founders developers (n=102).

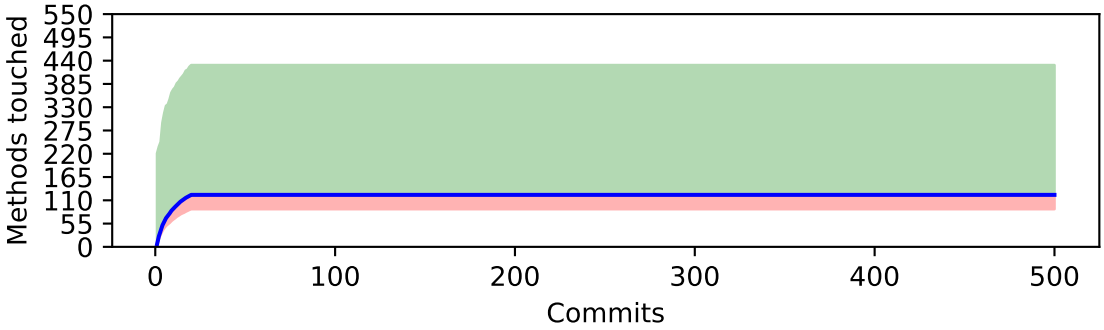
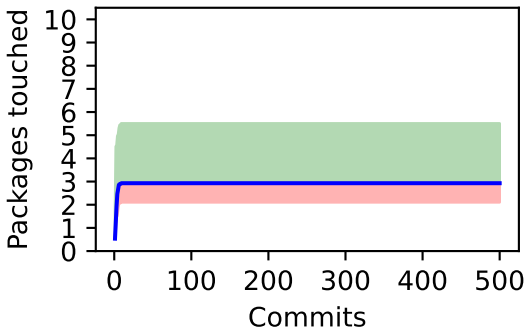
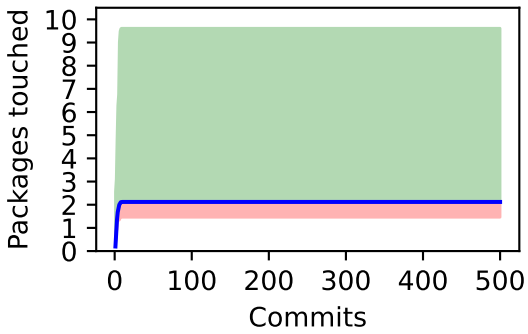


Figure 57: A time series of the average (mean) total methods touched (y-axis) against the number of commitss (x-axis) for All Joiners developers (n=2177).

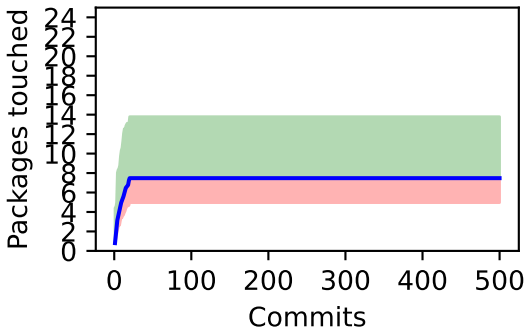
1.6.8 Time series components touched - For packages touched for each week commit



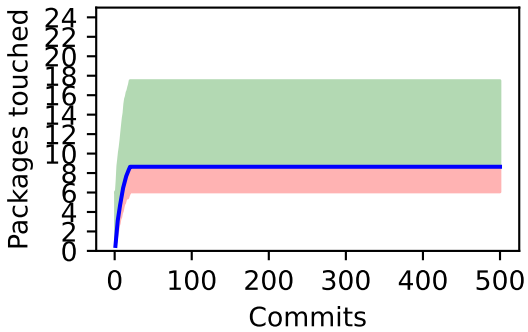
(a) Transient founder (n=28 & μ=2.93)



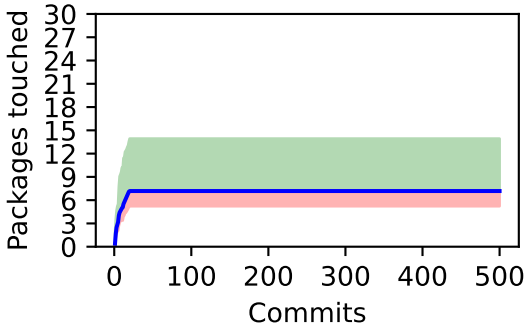
(b) Transient later joiner (n=1396 & μ=2.12)



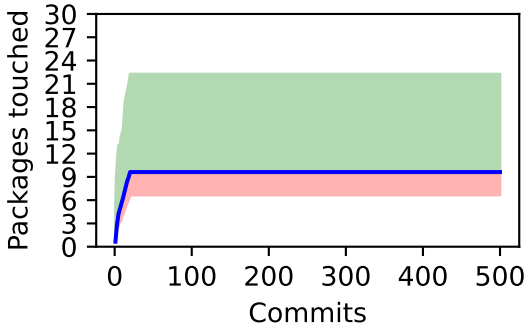
(c) Moderate founder (n=40 & μ=7.48)



(d) Moderate later joiner (n=580 & μ=8.65)

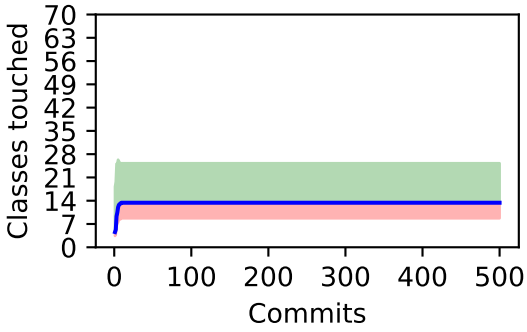


(e) Sustained founder (n=34 & μ=7.18)

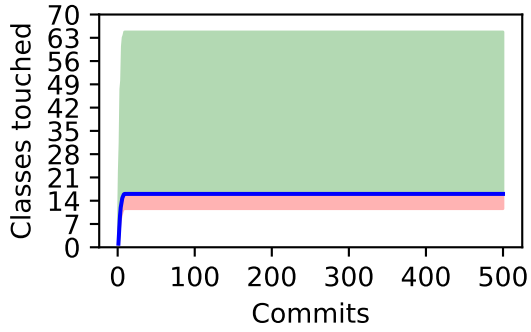


(f) Sustained later joiner (n=203 & μ=9.62)

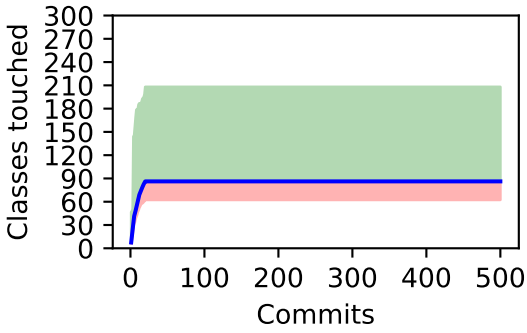
Figure 58: A time series of the average (mean) total packages touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



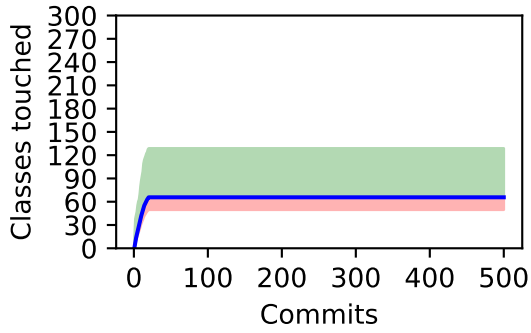
(a) Transient founder (n=28 & μ=13.39)



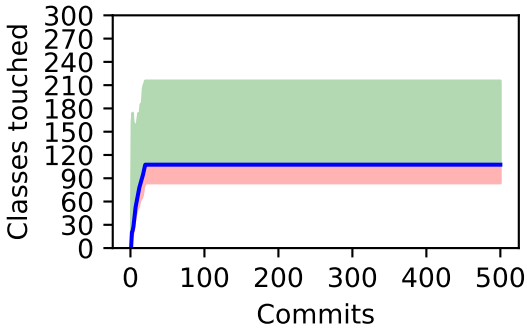
(b) Transient later joiner (n=1396 & μ=16.05)



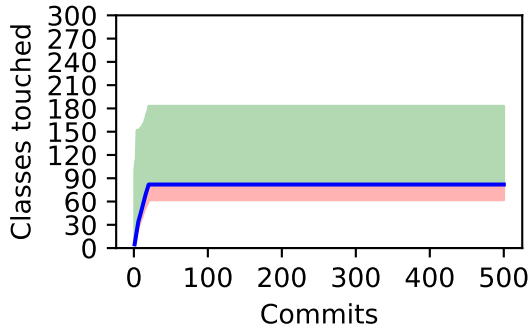
(c) Moderate founder (n=40 & μ=86.22)



(d) Moderate later joiner (n=580 & μ=65.7)

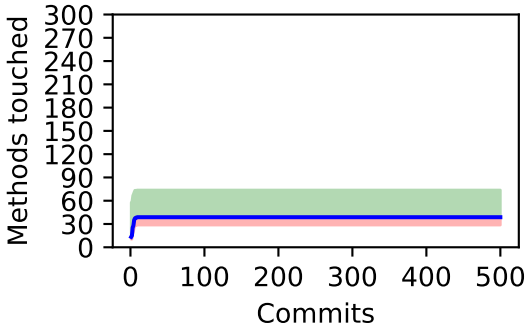


(e) Sustained founder (n=34 & μ=107.5)

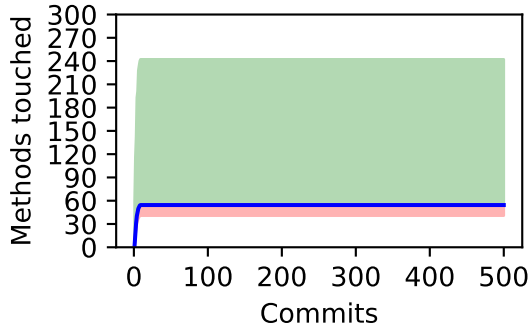


(f) Sustained later joiner (n=203 & μ=81.91)

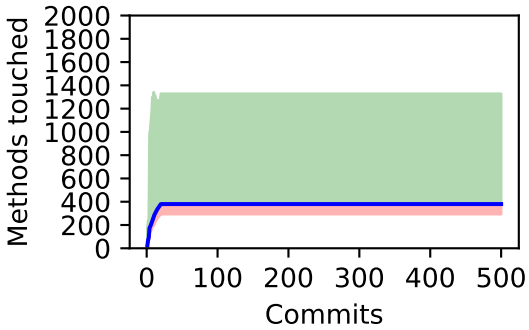
Figure 59: A time series of the average (mean) total classes touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.



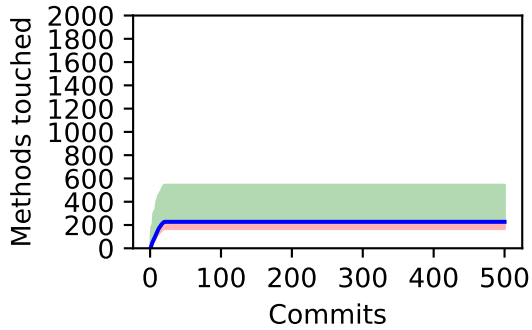
(a) Transient founder (n=28 & μ=38.75)



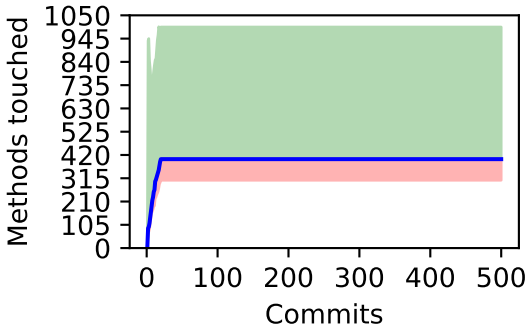
(b) Transient later joiner (n=1396 & μ=54.43)



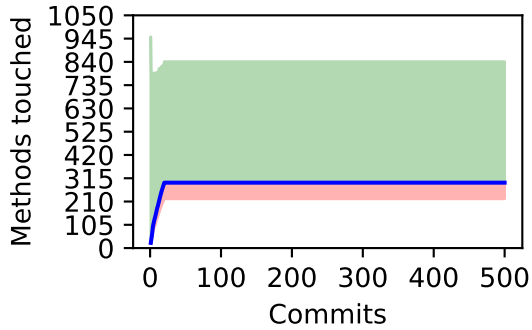
(c) Moderate founder (n=40 & μ=379.9)



(d) Moderate later joiner (n=580 & μ=227.38)



(e) Sustained founder (n=34 & μ=401.24)



(f) Sustained later joiner (n=203 & μ=295.1)

Figure 60: A time series of the average (mean) total methods touched (y-axis) against the number of commits (x-axis) for six (6) categories of developer.

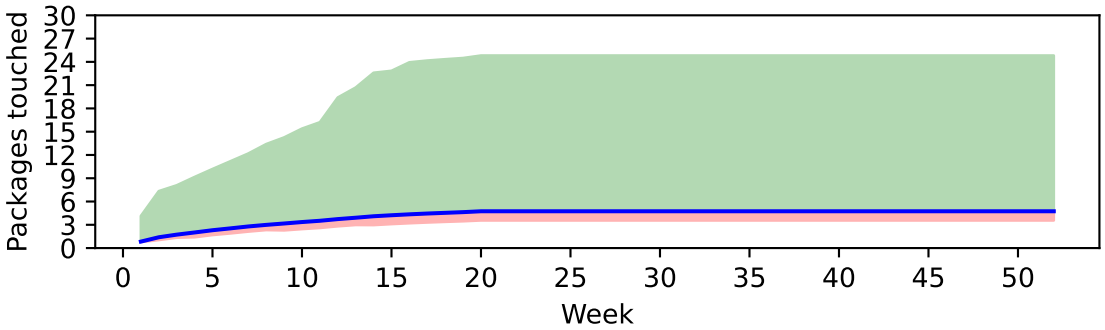


Figure 61: A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for All developers (n=2278).

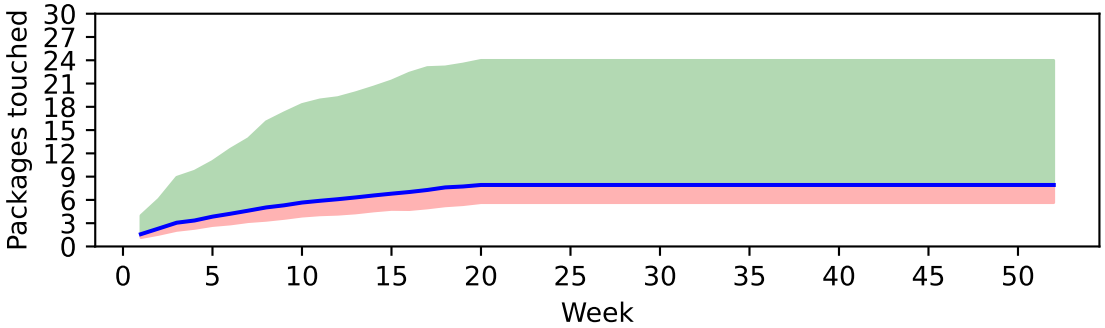


Figure 62: A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for All Founders developers (n=102).

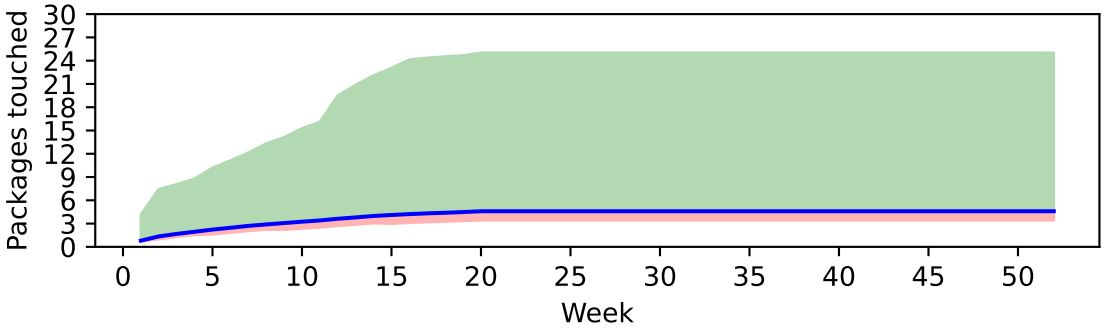


Figure 63: A time series of the average (mean) total packages touched (y-axis) against the number of weeks (x-axis) for All Joiners developers (n=2177).

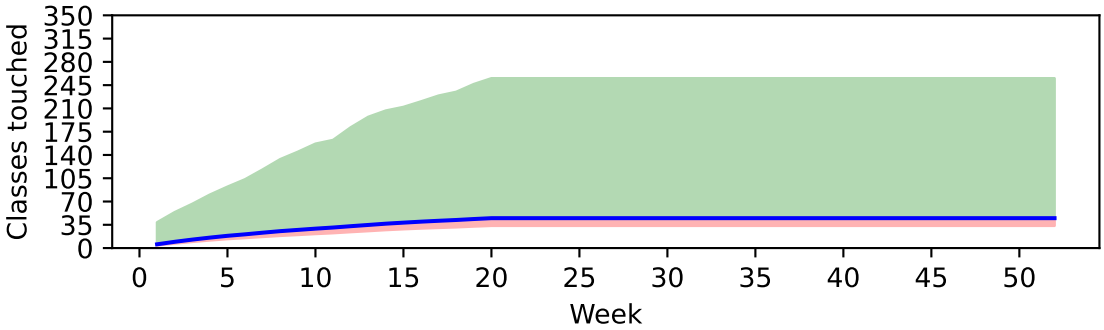


Figure 64: A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for All developers (n=2278).

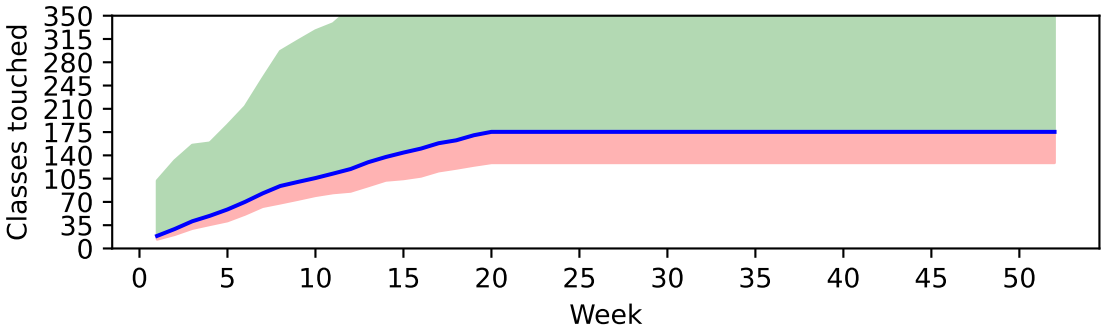


Figure 65: A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for All Founders developers (n=102).

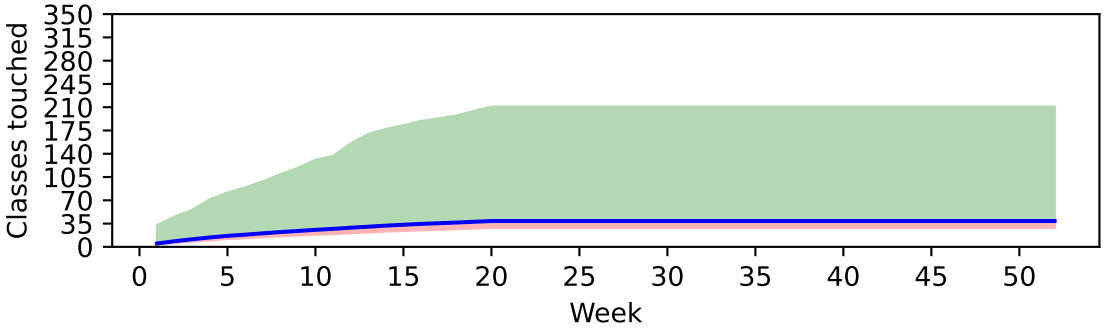


Figure 66: A time series of the average (mean) total classes touched (y-axis) against the number of weeks (x-axis) for All Joiners developers (n=2177).

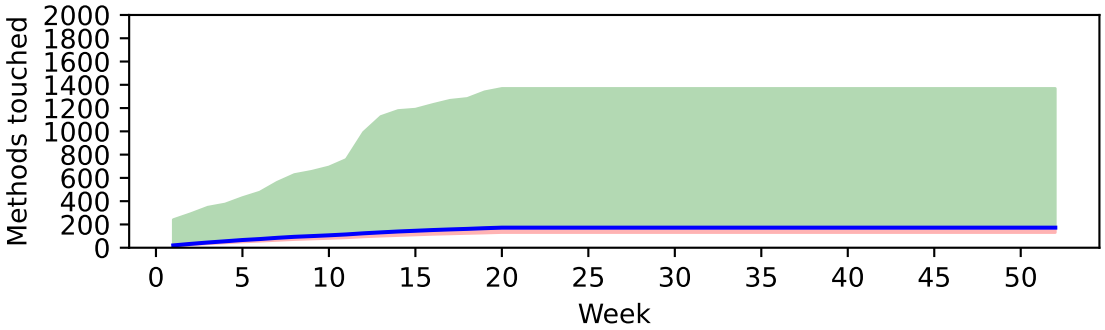


Figure 67: A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for All developers (n=2278).

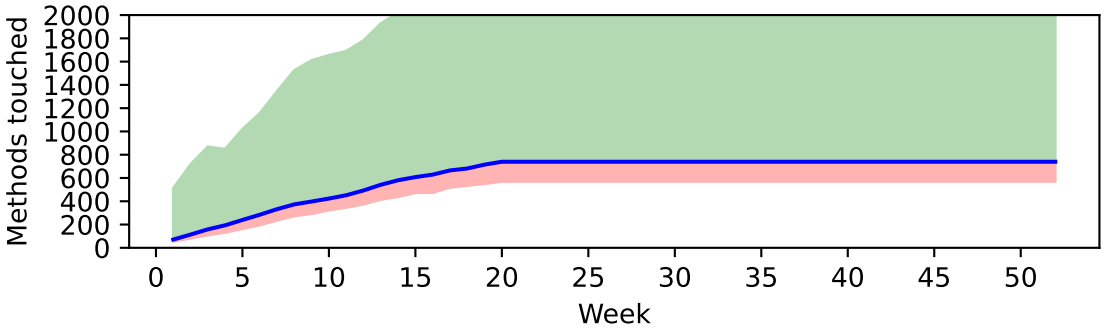


Figure 68: A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for All Founders developers (n=102).

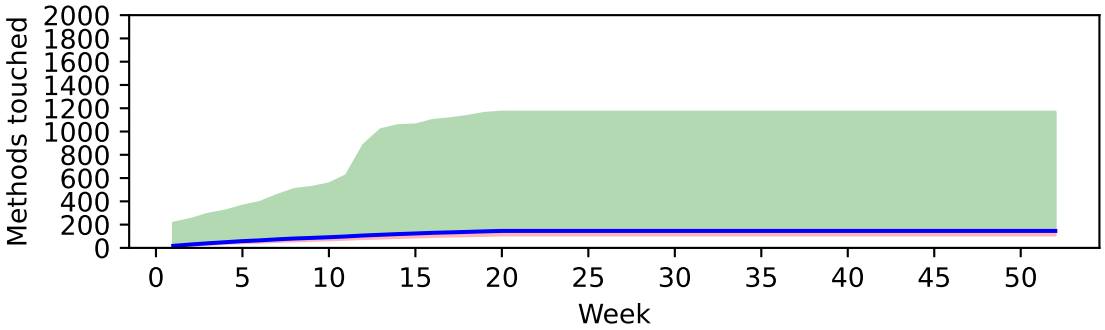
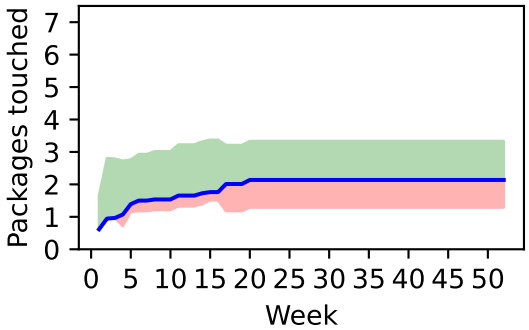
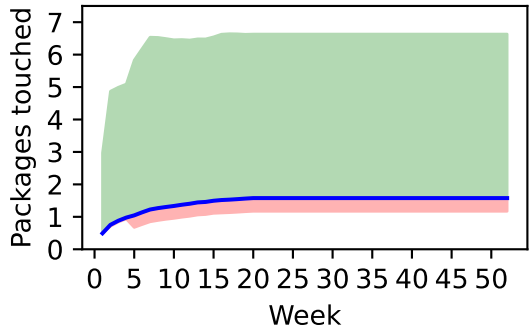


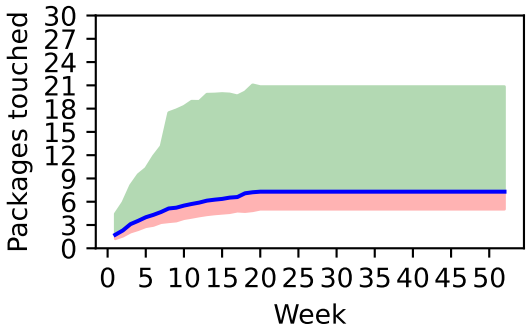
Figure 69: A time series of the average (mean) total methods touched (y-axis) against the number of weeks (x-axis) for All Joiners developers (n=2177).



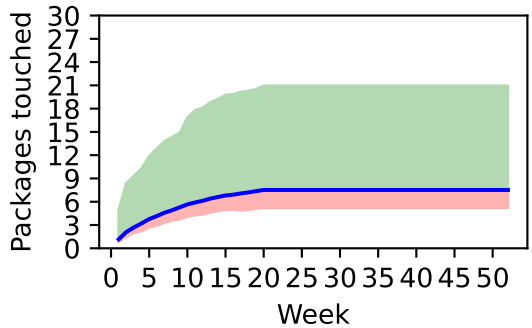
(a) Transient founder (n=28 & μ=2.14)



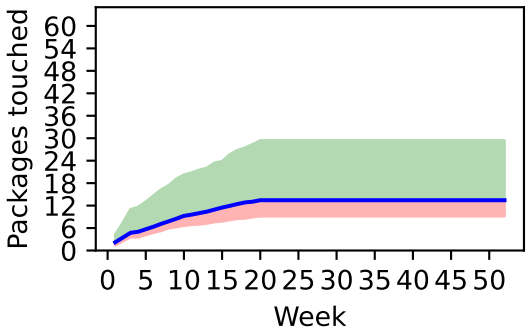
(b) Transient later joiner (n=1396 & μ=1.58)



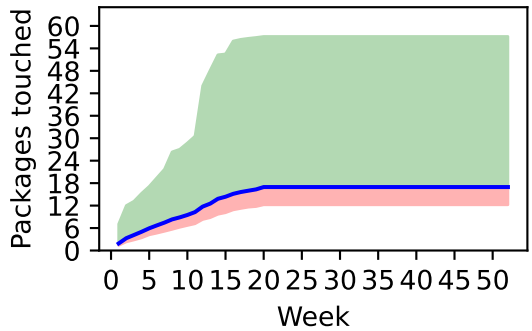
(c) Moderate founder (n=40 & μ=7.3)



(d) Moderate later joiner (n=580 & μ=7.52)

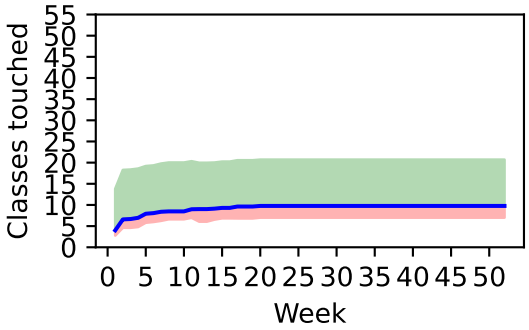


(e) Sustained founder (n=34 & μ=13.44)

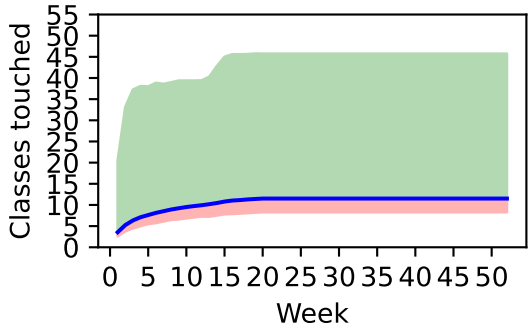


(f) Sustained later joiner (n=203 & μ=16.96)

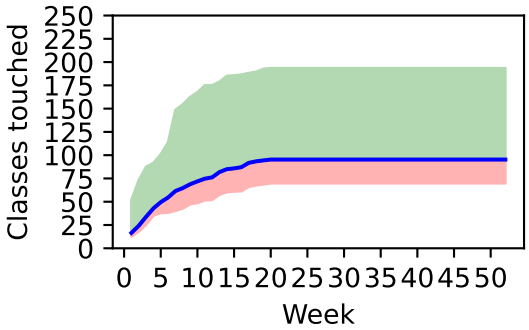
Figure 70: A time series of the average (mean) total packages touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.



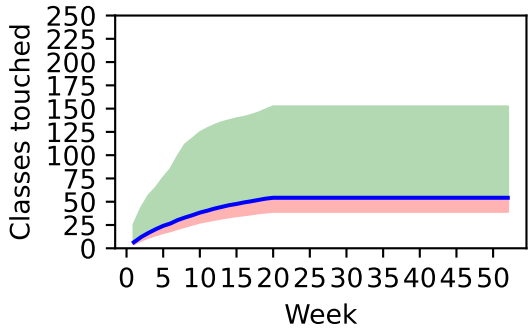
(a) Transient founder (n=28 & μ=9.77)



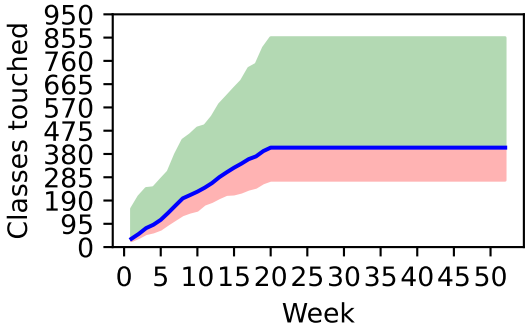
(b) Transient later joiner (n=1396 & μ=11.51)



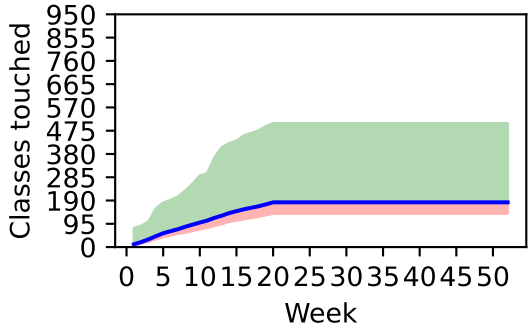
(c) Moderate founder (n=40 & μ=95.29)



(d) Moderate later joiner (n=580 & μ=54.26)

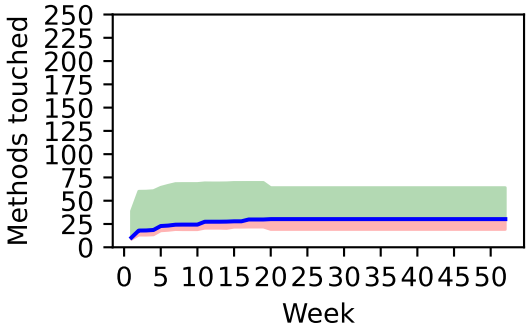


(e) Sustained founder (n=34 & μ=405.9)

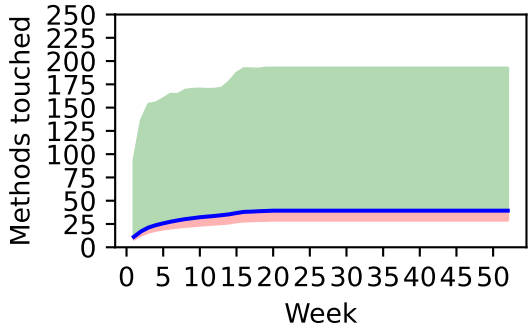


(f) Sustained later joiner (n=203 & μ=182.63)

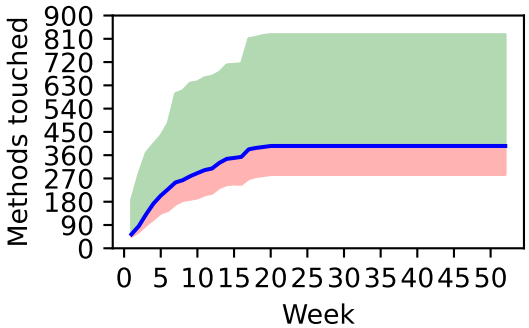
Figure 71: A time series of the average (mean) total classes touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.



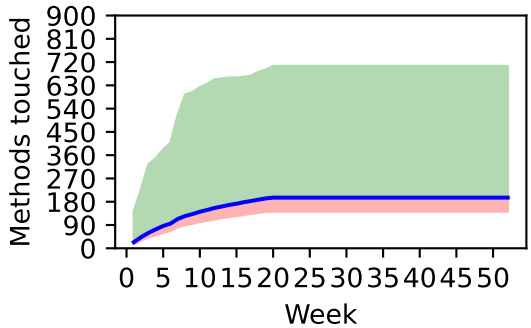
(a) Transient founder (n=28 & μ=30.24)



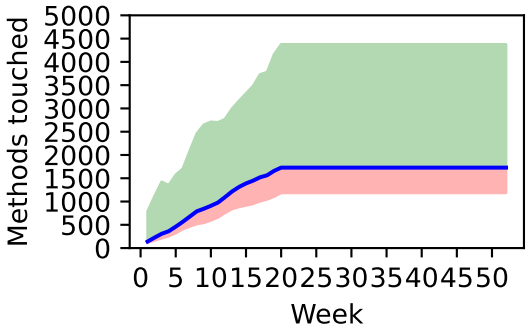
(b) Transient later joiner (n=1396 & μ=39.31)



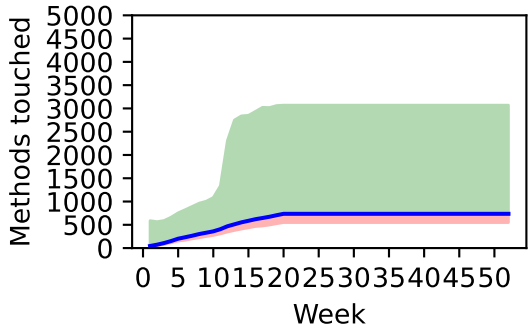
(c) Moderate founder (n=40 & μ=395.58)



(d) Moderate later joiner (n=580 & μ=195.87)



(e) Sustained founder (n=34 & μ=1728.34)



(f) Sustained later joiner (n=203 & μ=736.19)

Figure 72: A time series of the average (mean) total methods touched (y-axis) against the number of week (x-axis) for six (6) categories of developer.

1.7 Welch t test - For all components touched for all commits by Founder against Late joiner

A Welch t-test and Mann-Whitney U Test for all components touched on average.

Table 2: Welch t-test and Mann-Whitney U Results for Components and number of commits by Founder against Late joiner

Number of First Commits	**Component**	**Founder μ**	**Late joiner μ**	**Welch Statistic**	**Welch P**	**Mann-Whitney U Statistic**	**Mann-Whitney U P**
5	Packages	13.37	6.88	3.08	0.0026	141712.0	0.0000
10	Packages	13.37	6.88	3.08	0.0026	141712.0	0.0000
20	Packages	13.37	6.88	3.08	0.0026	141712.0	0.0000
5	Classes	309.92	74.68	5.08	0.0000	158119.5	0.0000
10	Classes	309.92	74.68	5.08	0.0000	158119.5	0.0000
20	Classes	309.92	74.68	5.08	0.0000	158119.5	0.0000
5	Methods	1284.91	273.68	4.75	0.0000	158996.5	0.0000
10	Methods	1284.91	273.68	4.75	0.0000	158996.5	0.0000
20	Methods	1284.91	273.68	4.75	0.0000	158996.5	0.0000

1.8 Welch t test - For all components touched for all commits by Moderate against Sustained

A Welch t-test and Mann-Whitney U Test for all components touched on average.

Table 3: Welch t-test and Mann-Whitney U Results for Components and number of commits by Moderate against Sustained

Number of First Commits	**Component**	**Moderate μ**	**Sustained μ**	**Welch Statistic**	**Welch P**	**Mann-Whitney U Statistic**	**Mann-Whitney U P**
5	Packages	10.99	5.74	6.95	0.0000	659794.5	0.0000
10	Packages	10.99	5.74	6.95	0.0000	659794.5	0.0000
20	Packages	10.99	5.74	6.95	0.0000	659794.5	0.0000
5	Classes	90.95	83.06	1.03	0.3033	819217.5	0.0000
10	Classes	90.95	83.06	1.03	0.3033	819217.5	0.0000
20	Classes	90.95	83.06	1.03	0.3033	819217.5	0.0000
5	Methods	331.48	314.20	0.53	0.5939	805485.0	0.0000
10	Methods	331.48	314.20	0.53	0.5939	805485.0	0.0000
20	Methods	331.48	314.20	0.53	0.5939	805485.0	0.0000

Statistical Models

Welch's t-test To specifically compare the two primary roles (**Sample A** vs. **Sample B**) without assuming equal variances or sample sizes, Welch's t-test is employed. The t-statistic is defined as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}} \tag{1}$$

Where:

- $\bar{X}_1, \bar{X}_2$ are the sample means of Sample A and Sample B.
- s_1^2, s_2^2 are the sample variances.
- N_1, N_2 are the sample sizes.

The degrees of freedom (ν) for this test are calculated using the Welch-Satterthwaite equation:

$$\nu \approx \frac{\left(\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}\right)^2}{\frac{(s_1^2/N_1)^2}{N_1-1} + \frac{(s_2^2/N_2)^2}{N_2-1}} \tag{2}$$

1.9 Anova generate - For all components touched for first five commits

An Anova table for all components touched on average.

1.9.1 Anova generate - For packages touched for a first five commits

An Anova table for packages touched on average.

Table 4: Two-Way ANOVA for packages touched by Developer Category for first five commits

	Sum Sq	df	F	p-Value
Intercept	497.0250	1.0000	25.6264	0.0000
C(Role)	1.4330	1.0000	0.0739	0.7858
C(Tenure)	12.8219	2.0000	0.3305	0.7186
C(Role):C(Tenure)	47.7037	2.0000	1.2298	0.2925
Residual	44123.7544	2275.0000	NaN	NaN

1.9.2 Anova generate - For classes touched for a first five commits

An Anova table for classes touched on average.

Table 5: Two-Way ANOVA for classes touched by Developer Category for first five commits

	Sum Sq	df	F	p-Value
Intercept	68475.6250	1.0000	64.1354	0.0000
C(Role)	13981.9502	1.0000	13.0957	0.0003
C(Tenure)	15297.7980	2.0000	7.1641	0.0008
C(Role):C(Tenure)	7200.0876	2.0000	3.3719	0.0345
Residual	2428957.5236	2275.0000	NaN	NaN

1.9.3 Anova generate - For methods touched for a first five commits

An Anova table for methods touched on average.

Table 6: Two-Way ANOVA for methods touched by Developer Category for first five commits

	Sum Sq	df	F	p-Value
Intercept	1395022.5000	1.0000	53.0497	0.0000
C(Role)	451979.7037	1.0000	17.1878	0.0000
C(Tenure)	383522.0154	2.0000	7.2923	0.0007
C(Role):C(Tenure)	254925.9061	2.0000	4.8471	0.0079
Residual	59824573.4495	2275.0000	NaN	NaN

1.9.4 Anova generate - For packages touched for a first ten commits

An Anova table for packages touched on average.

1.9.5 Anova generate - For classes touched for a first ten commits

An Anova table for classes touched on average.

Table 7: Two-Way ANOVA for packages touched by Developer Category for first ten commits

	Sum Sq	df	F	p-Value
Intercept	1060.9000	1.0000	29.6901	0.0000
C(Role)	33.0395	1.0000	0.9246	0.3364
C(Tenure)	93.5331	2.0000	1.3088	0.2703
C(Role):C(Tenure)	60.1476	2.0000	0.8416	0.4311
Residual	81291.2154	2275.0000	NaN	NaN

Table 8: Two-Way ANOVA for classes touched by Developer Category for first ten commits

	Sum Sq	df	F	p-Value
Intercept	151782.4000	1.0000	91.2065	0.0000
C(Role)	14448.4548	1.0000	8.6821	0.0032
C(Tenure)	54603.0646	2.0000	16.4056	0.0000
C(Role):C(Tenure)	10049.5659	2.0000	3.0194	0.0490
Residual	3785966.8647	2275.0000	NaN	NaN

1.9.6 Anova generate - For methods touched for a first ten commits

An Anova table for methods touched on average.

Table 9: Two-Way ANOVA for methods touched by Developer Category for first ten commits

	Sum Sq	df	F	p-Value
Intercept	2946861.2250	1.0000	77.7203	0.0000
C(Role)	596144.3889	1.0000	15.7227	0.0001
C(Tenure)	1029687.0534	2.0000	13.5784	0.0000
C(Role):C(Tenure)	319730.2008	2.0000	4.2163	0.0149
Residual	86259457.2963	2275.0000	NaN	NaN

1.9.7 Anova generate - For packages touched for a first twenty commits

An Anova table for packages touched on average.

Table 10: Two-Way ANOVA for packages touched by Developer Category for first twenty commits

	Sum Sq	df	F	p-Value
Intercept	2235.0250	1.0000	38.1943	0.0000
C(Role)	51.3593	1.0000	0.8777	0.3489
C(Tenure)	396.5698	2.0000	3.3885	0.0339
C(Role):C(Tenure)	151.5806	2.0000	1.2952	0.2741
Residual	133126.8918	2275.0000	NaN	NaN

1.9.8 Anova generate - For classes touched for a first twenty commits

An Anova table for classes touched on average.

Table 11: Two-Way ANOVA for classes touched by Developer Category for first twenty commits

	Sum Sq	df	F	p-Value
Intercept	297390.0250	1.0000	129.0043	0.0000
C(Role)	15761.2138	1.0000	6.8370	0.0090
C(Tenure)	146938.1700	2.0000	31.8700	0.0000
C(Role):C(Tenure)	12952.2768	2.0000	2.8093	0.0605
Residual	5244492.8354	2275.0000	NaN	NaN

1.9.9 Anova generate - For methods touched for a first twenty commits

An Anova table for methods touched on average.

Table 12: Two-Way ANOVA for methods touched by Developer Category for first twenty commits

	Sum Sq	df	F	p-Value
Intercept	5772960.4000	1.0000	118.7808	0.0000
C(Role)	870509.3766	1.0000	17.9111	0.0000
C(Tenure)	2510362.6108	2.0000	25.8258	0.0000
C(Role):C(Tenure)	460313.5876	2.0000	4.7356	0.0089
Residual	110569092.3114	2275.0000	NaN	NaN

1.9.10 Anova generate - For packages touched for a all commits

An Anova table for packages touched on average.

Table 13: Two-Way ANOVA for packages touched by Developer Category for all commits

	Sum Sq	df	F	p-Value
Intercept	2235.0250	1.0000	38.1943	0.0000
C(Role)	51.3593	1.0000	0.8777	0.3489
C(Tenure)	396.5698	2.0000	3.3885	0.0339
C(Role):C(Tenure)	151.5806	2.0000	1.2952	0.2741
Residual	133126.8918	2275.0000	NaN	NaN

1.9.11 Anova generate - For classes touched for a all commits

An Anova table for classes touched on average.

Table 14: Two-Way ANOVA for classes touched by Developer Category for all commits

	Sum Sq	df	F	p-Value
Intercept	297390.0250	1.0000	129.0043	0.0000
C(Role)	15761.2138	1.0000	6.8370	0.0090
C(Tenure)	146938.1700	2.0000	31.8700	0.0000
C(Role):C(Tenure)	12952.2768	2.0000	2.8093	0.0605
Residual	5244492.8354	2275.0000	NaN	NaN

1.9.12 Anova generate - For methods touched for a all commits

An Anova table for methods touched on average.

Table 15: Two-Way ANOVA for methods touched by Developer Category for all commits

	Sum Sq	df	F	p-Value
Intercept	5772960.4000	1.0000	118.7808	0.0000
C(Role)	870509.3766	1.0000	17.9111	0.0000
C(Tenure)	2510362.6108	2.0000	25.8258	0.0000
C(Role):C(Tenure)	460313.5876	2.0000	4.7356	0.0089
Residual	110569092.3114	2275.0000	NaN	NaN

Statistical Model

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric (Y_{ijk}) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{3}$$

Where:

- μ is the grand mean of the touched components.
- α_i represents the effect of the **Role** (Founder vs. Late Joiner).
- β_j represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- ϵ_{ijk} is the residual error.

The F-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{4}$$

1.10 Strategy logistic regression - For all components touched for all commits

A Strategy Logistic Regression table for all components touched on average.

Table 16: Strategy Logistic Regression for all components against number of commits

Number of First Commits	Component	Type	Coefficient	Odds Ratio	Std. Err	p-Value
5	packages	Intercept	-2.3271	0.0976	0.0804	0.0000
5	packages	Packages Count	0.0561	1.0578	0.0118	0.0000
10	packages	Intercept	-2.3483	0.0955	0.0817	0.0000
10	packages	Packages Count	0.0451	1.0461	0.0088	0.0000
20	packages	Intercept	-2.4687	0.0847	0.0848	0.0000
20	packages	Packages Count	0.0514	1.0527	0.0064	0.0000
All	packages	Intercept	-2.4687	0.0847	0.0848	0.0000
All	packages	Packages Count	0.0514	1.0527	0.0064	0.0000
5	classes	Intercept	-2.3183	0.0984	0.0781	0.0000
5	classes	Classes Count	0.0077	1.0077	0.0016	0.0000
10	classes	Intercept	-2.4435	0.0869	0.0837	0.0000
10	classes	Classes Count	0.0087	1.0087	0.0012	0.0000
20	classes	Intercept	-2.7221	0.0657	0.0946	0.0000
20	classes	Classes Count	0.0115	1.0116	0.0011	0.0000
All	classes	Intercept	-2.7221	0.0657	0.0946	0.0000
All	classes	Classes Count	0.0115	1.0116	0.0011	0.0000
5	methods	Intercept	-2.2561	0.1048	0.0742	0.0000
5	methods	Methods Count	0.0013	1.0013	0.0003	0.0000
10	methods	Intercept	-2.3485	0.0955	0.0789	0.0000
10	methods	Methods Count	0.0016	1.0016	0.0003	0.0000
20	methods	Intercept	-2.5481	0.0782	0.0866	0.0000
20	methods	Methods Count	0.0023	1.0023	0.0003	0.0000
All	methods	Intercept	-2.5481	0.0782	0.0866	0.0000
All	methods	Methods Count	0.0023	1.0023	0.0003	0.0000

Predictive Identification Model

To determine if early-stage developer behavior can identify future sustained contributors, we employ a Logistic Regression model. The probability P of a developer becoming a **Sustained Contributor** is modeled using the log-odds (logit) function:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \epsilon \tag{5}$$

Where:

- P is the probability that the developer's tenure is "Sustained".
- β_0 is the intercept.
- X_n represents early-stage metrics (e.g., methods or classes touched in the first 10, 20, or 50 commits).
- β_n are the coefficients representing the weight of each predictor.

The impact of each behavior is quantified using the **Odds Ratio (OR)**, calculated as:

$$OR = e^{\beta_n} \tag{6}$$

An $OR > 1$ indicates that an increase in the metric (e.g., touching more classes) increases the likelihood of long-term retention.

Model Evaluation The predictive power is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). An AUC of 0.5 represents random chance, while an AUC closer to 1.0 indicates high predictive accuracy.

2 Repository: 14

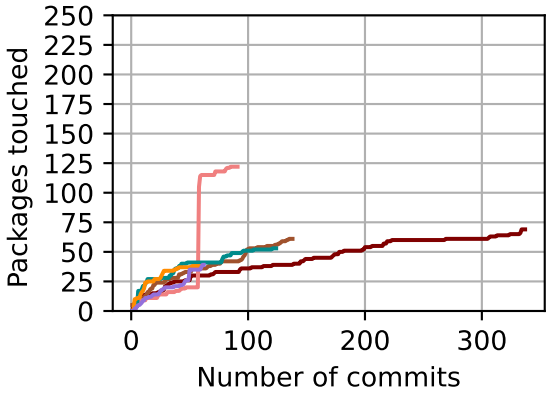
2.1 Time series sustained components touched by developer - 14 For developer commits and components touched.

Table 17: Table of first 6 sustained founder developers in the Scatter Developer graphs.

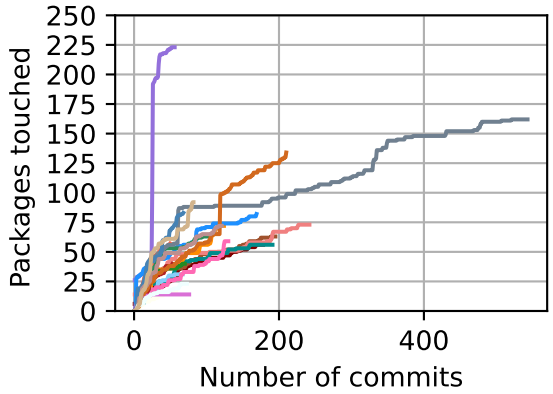
Developer ID	Colour
2930	Maroon
2932	Sienna
2934	LightCoral
2935	DarkCyan
2946	DarkOrange
2959	MediumPurple

Table 18: Table of first 21 sustained later joiner developers in the Scatter Developer graphs.

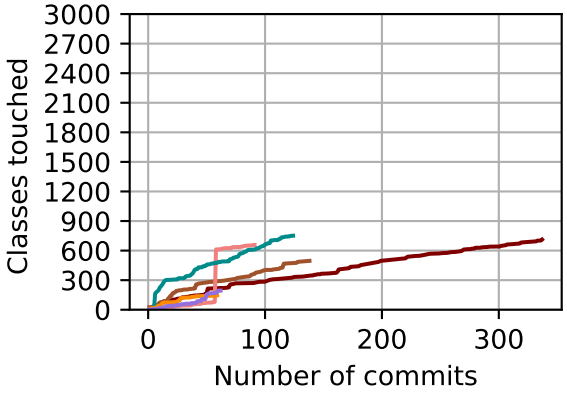
Developer ID	Colour
1774	Maroon
2986	Sienna
3000	LightCoral
3020	DarkCyan
3037	DarkOrange
3050	MediumPurple
3071	LightSkyBlue
3072	HotPink
3141	ForestGreen
3143	FireBrick
3163	DodgerBlue
3172	Tomato
3178	SeaGreen
3216	SlateGray
3217	Ivory
3272	Orchid
3339	Chocolate
3354	SteelBlue
3456	RosyBrown
3473	Azure
3485	Tan



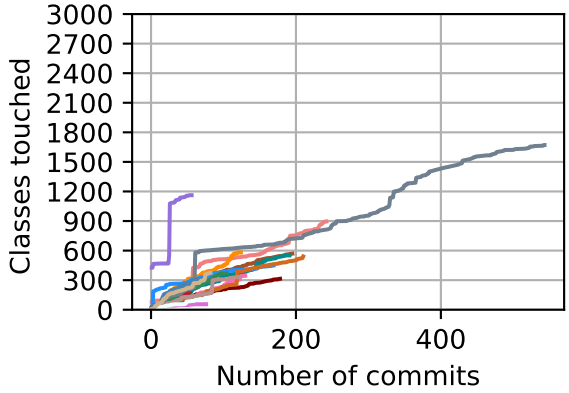
(a) Packages for sustained founder



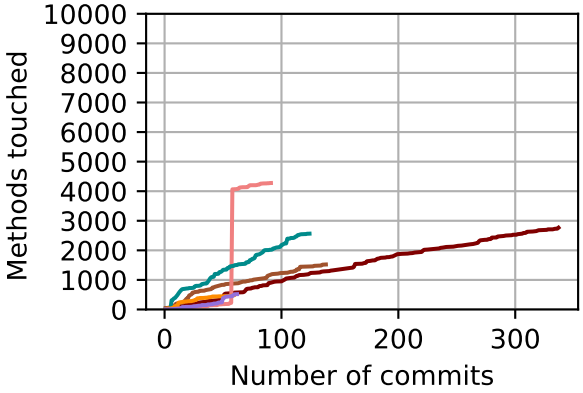
(b) Packages for sustained later joiner



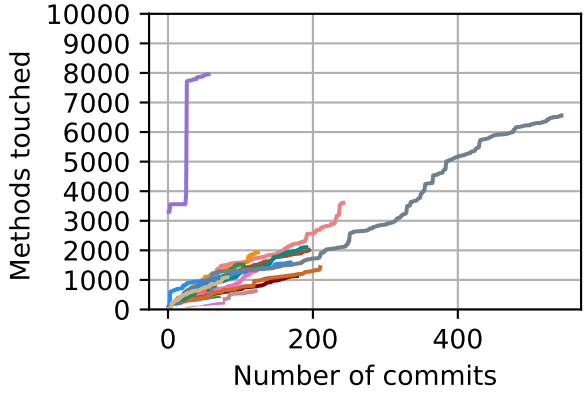
(c) Classes for sustained founder



(d) Classes for sustained later joiner



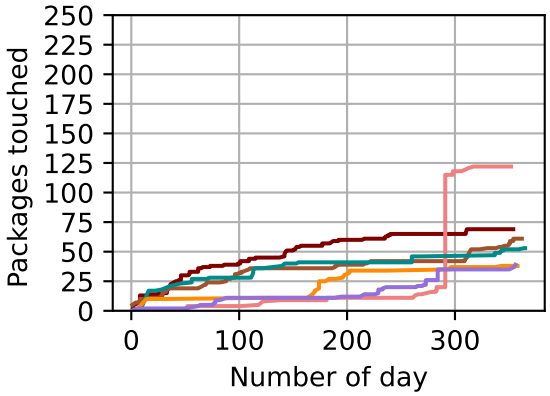
(e) Methods for sustained founder



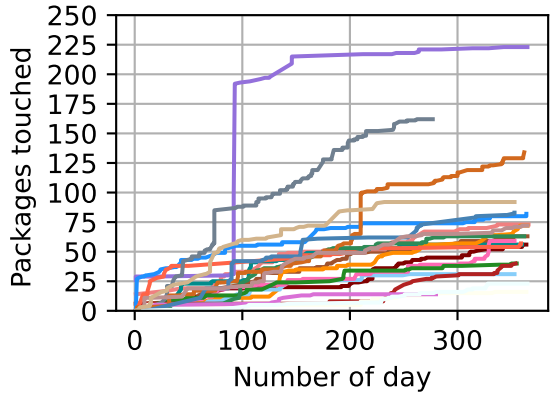
(f) Methods for sustained later joiner

Figure 73: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 17. Sustained later joiner developers with colours are shown in Table 18.

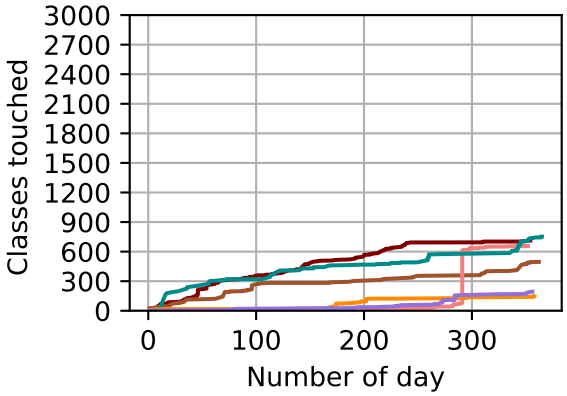
2.2 Time series sustained components touched by developer - 14 For developer day and components touched.



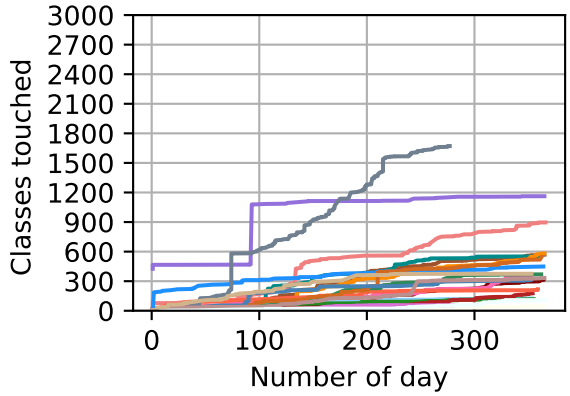
(a) Packages for sustained founder



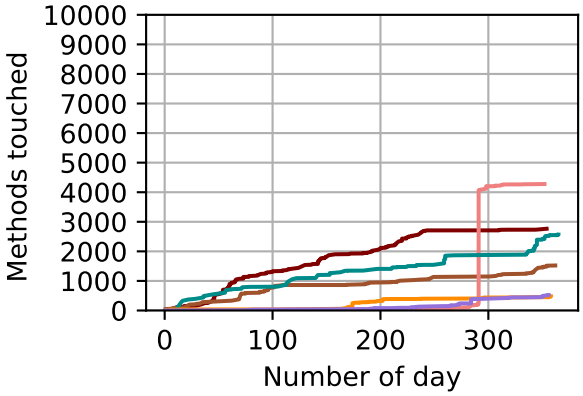
(b) Packages for sustained later joiner



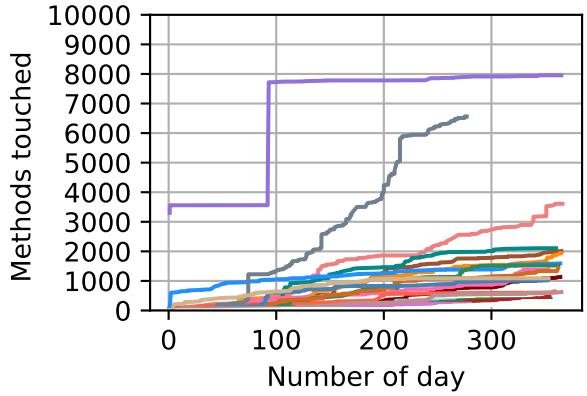
(c) Classes for sustained founder



(d) Classes for sustained later joiner



(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 74: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 17. Sustained later joiner developers with colours are shown in Table 18.

3 Repository: 21

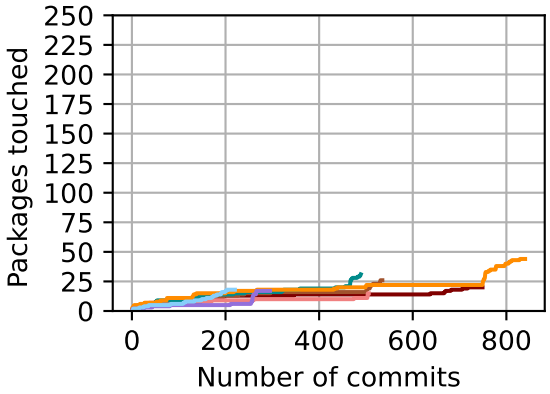
3.1 Time series sustained components touched by developer - 21 For developer commits and components touched.

Table 19: Table of first 7 sustained founder developers in the Scatter Developer graphs.

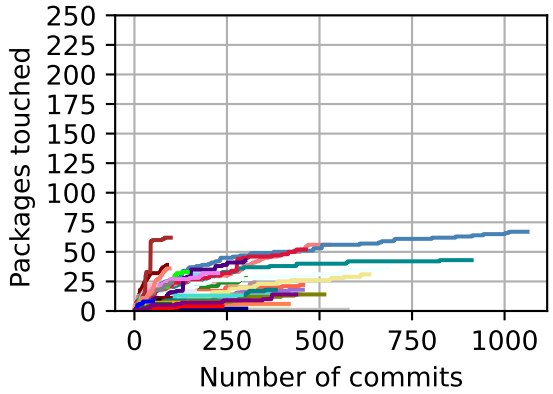
Developer ID	Colour
6245	Maroon
6246	Sienna
6247	LightCoral
6249	DarkCyan
6250	DarkOrange
6251	MediumPurple
6257	LightSkyBlue

Table 20: Table of first 50 sustained later joiner developers in the Scatter Developer graphs.

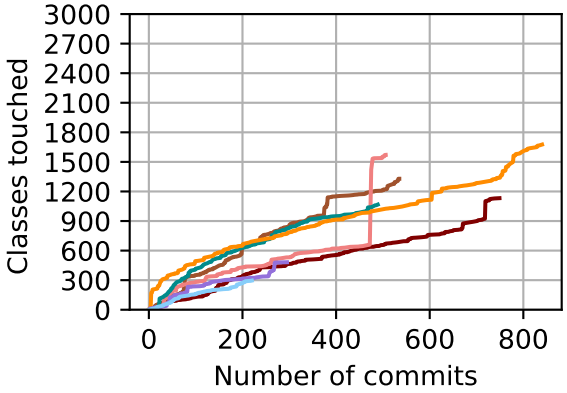
Developer ID	Colour
199	Maroon
6252	Sienna
6253	LightCoral
6259	DarkCyan
6260	DarkOrange
6262	MediumPurple
6263	LightSkyBlue
6264	HotPink
6267	ForestGreen
6271	FireBrick
6272	DodgerBlue
6276	Tomato
6278	SeaGreen
6280	SlateGray
6282	Ivory
6284	Orchid
6288	Chocolate
6295	SteelBlue
6297	RosyBrown
6298	Azure
6302	Tan
6306	Plum
6309	Crimson
6312	Khaki
6313	Lavender
6314	Indigo
6315	Turquoise
6326	Beige
6329	Silver
6333	Gold
6338	Coral
6342	Lime
6343	Maroon
6349	Navy
6352	Teal
6355	Olive
6363	Violet
6368	Gray
6369	Pink
6370	Brown
6373	Black
6374	Yellow
6375	Magenta
6379	Cyan
6391	Salmon
6410	Orange
6411	Red
6417	Green
6418	Blue
6420	Purple



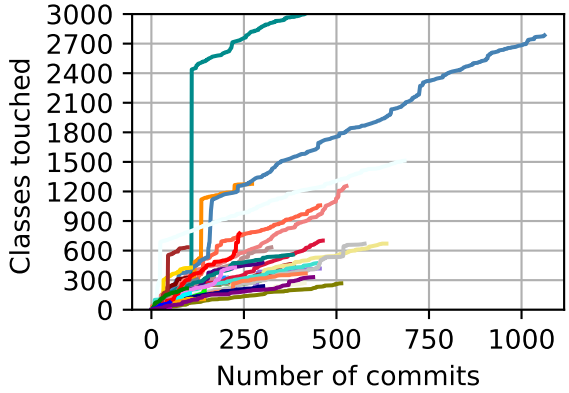
(a) Packages for sustained founder



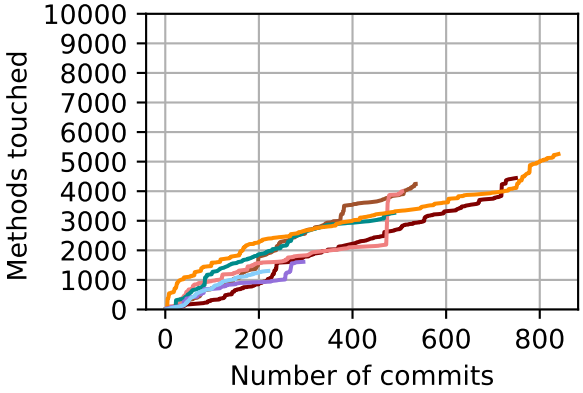
(b) Packages for sustained later joiner



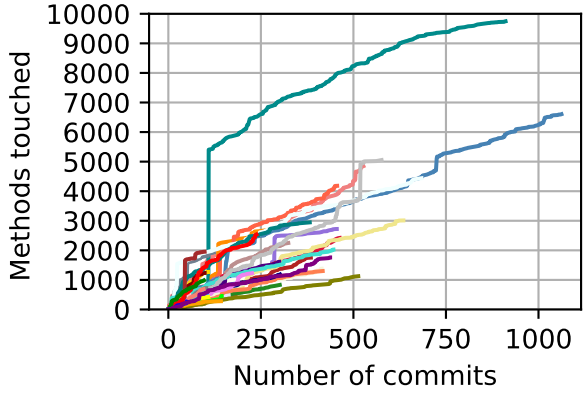
(c) Classes for sustained founder



(d) Classes for sustained later joiner



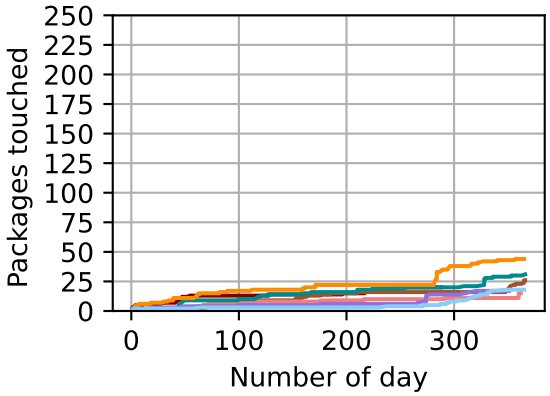
(e) Methods for sustained founder



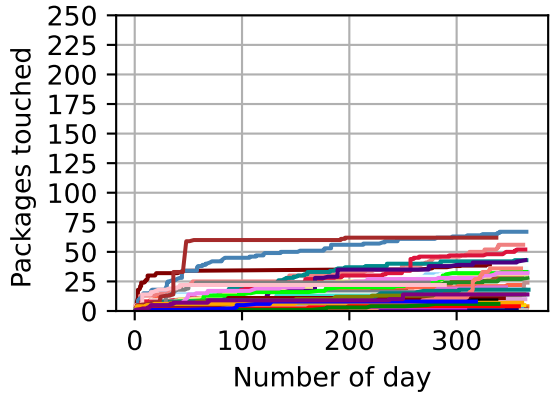
(f) Methods for sustained later joiner

Figure 75: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 19. Sustained later joiner developers with colours are shown in Table 20.

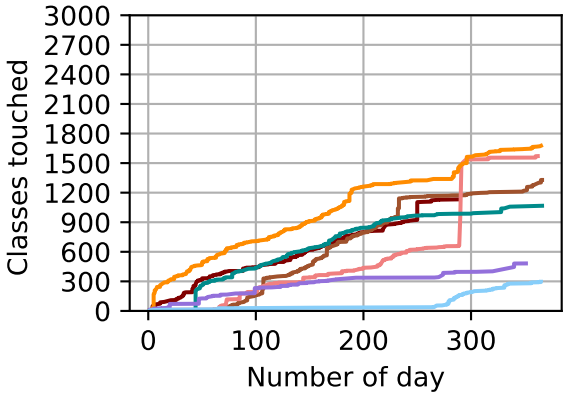
3.2 Time series sustained components touched by developer - 21 For developer day and components touched.



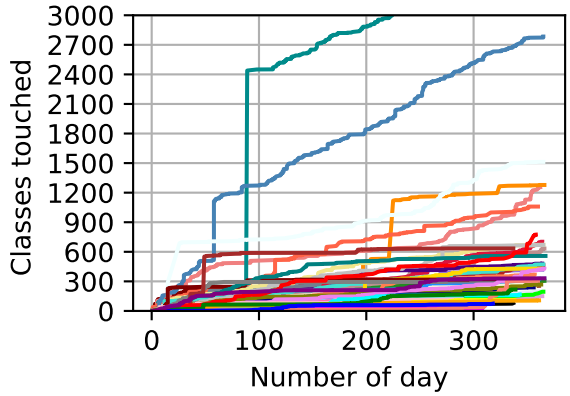
(a) Packages for sustained founder



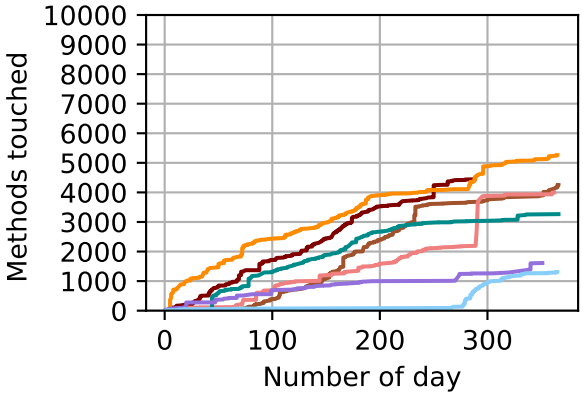
(b) Packages for sustained later joiner



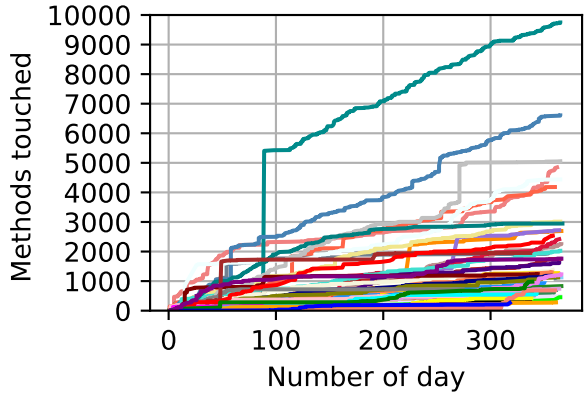
(c) Classes for sustained founder



(d) Classes for sustained later joiner



(e) Methods for sustained founder



(f) Methods for sustained later joiner

Figure 76: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 19. Sustained later joiner developers with colours are shown in Table 20.